

Party Competition on the Environment

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Abstract

Parties' policy preferences play important roles in shaping policy responses to environmental problems. The characteristics of the environment as a 'new' political issue have provoked considerable debate. Is the environment a valence or a positional issue? Does the cross-cutting nature of the environment make it independent of existing political conflict structures or have established parties incorporated it into those existing structures? This paper seeks to describe and explain variation in party preferences on the environment. It does this by analysing structures of competition on the environment at the system level and on the mechanisms that produce them at the party level. It makes three contributions: first, it contributes to debates about the nature of the environmental issue (the valence/positional debate) by providing a system-level description of the structures of competition on the environment, drawing on a range of data sources. Second, it revisits existing explanations for the politicisation of the environment, contributing to existing research on the roles of niche parties, public attitudes and environmental problems. Third, it tests a new argument, which suggests that the historic structure of conflict underpinning a party system shapes the politicisation of the environment, and it finds support for this argument.

Keywords (up to 5): political parties, environment, problem pressure, historic conflicts

Introduction

As a political issue, the environment is associated with major societal challenges such as climate change and the biodiversity crisis, and representative politics is central to societal responses to those challenges. Political parties shape government policies on the environment (e.g., Jahn, 2016; Knill et al., 2010; Leinaweaver & Thomson, 2016; Lundquist, 2024; Schulze, 2014) and they act as societal leaders, influencing the public's environmental attitudes (e.g., Carmichael & Brulle, 2017; Egan & Mullin, 2017; Linde, 2020). Indeed, it is plausible that major environmental problems cannot be mitigated effectively without the engagement of political parties and their leaders.

This article focuses on two aspects of the party politics of the environment: the politicisation of the environment and the 'greening' of party systems. The first step in the politicisation of the environment came with the emergence of the issue onto the policy agenda, driven by public, civil society and political engagement (Downs 1972; McCormick 1989). However, we know little about whether the environment has continued to be increasingly politicised in party systems since then. We ask to what extent and in what ways has the environment been politicised. Has the issue become more salient and more polarised over time? Our second question is whether politics has been progressively 'greened' over time, through the adoption of pro-environmental policies and priorities, and the increasing success of environmentalist parties. Have party systems become 'greener' over time, characterised by more pro-environmental preferences?

We then examine differences in environmental politics between countries and over time, focusing on two explanations that have until now received little attention. The first explanation concerns environmental problem-pressure, which we expect to be associated with greater politicisation of the environment and with greener policy preferences. The second is an argument about how the strength of the overall meta-conflict in a party system shapes the politicisation of 'new' issues by influencing the level of political uncertainty associated with their politicisation (Green-Pedersen & Little, 2024). We expect the strength of this meta-conflict to shape politicisation via a number of specific mechanisms, which we also test. In addition to interrogating these two central arguments, we revisit some more established lines of inquiry on parties' environmental policy preferences. We re-examine how public attitudes shape the party politics of the environment and the still-contested role of niche parties in shaping the party politics of the environment (Abou-Chadi, 2016; Green-Pedersen, 2019; Schwörer, 2024; Spoon et al., 2014).

To address these questions, we draw on three different sources of cross-national comparative data on parties' environmental policy preferences: manifesto data, expert surveys and data coded by

experts in collaboration with parties for a voting advice application. This allows us to examine the robustness of results across these data sources, and presents a substantially broader set of analyses than previous studies.

Surprisingly, there is no evidence of a secular increase in the *salience* of the environment since 1990; rather, it is best described as cyclical, with peaks and troughs. Party positions have become less pro-environmental over time by all three measures we use. In sum, there is no evidence that the average party system in the scope of our study has ‘greened’ over time. Although salience has not increased over time, we do find that another aspect of politicisation – polarisation – has occurred: parties have pulled further apart on the environment.

This describes the average party system, and there are of course important differences among party systems, which we seek to explain with analyses at the system level and then at the party level. We find limited evidence for the influence of environmental policy problems on politicisation (specifically, polarisation) and ‘greening’ (specifically, positions). The strength of the historic left-right conflict shapes the politicisation of the environment, and there is also evidence that public attitudes shape the politicisation and greening of party systems. We observe a striking absence of evidence across all data sources for Green party strength eliciting accommodative policy preferences from other parties, whereas parties of the radical right (RR) do appear to shape the party politics of the environment.

The study’s key contributions lie in providing a cross-national description of the party politics of the environment; examining the under-studied role of problem-pressure in shaping the party politics of the environment; providing a first comparative test of a new argument concerning the role of historic conflict structures in the politicisation of ‘new politics’ issues; adding to knowledge on existing explanations concerning public attitudes and niche parties; and examining each of these questions by drawing on three different sources of comparative data on parties’ environmental policy preferences.

The paper proceeds as follows. First, we review debates around the nature of the environment as an issue, its development over time, and how vote-, office- and policy-seeking can explain variation in parties’ environmental policy preferences. Second, we outline two key arguments: that party preferences on the environment are influenced by problem-pressures and that long-term primary conflict structures shape competition on ‘new’ issues like the environment. Third, we describe how we measure the dependent variables at the system level and party level drawing on existing cross-national data sources, how we measure our independent variables, and our modelling

approach. Fourth, the analysis is carried out in three steps: the description of politicisation and ‘greening’ at the system level, analyses at the system level, and analyses at the party level. We then discuss the implications of the results and, finally, we identify avenues for future research.

Political parties and the environment

The characteristics of the environment as a political issue – and thus the nature of political competition on the environment – have provoked important debates. One concerns the nature of the environment as a political issue. The environment was commonly identified as a classic valence issue, where there is a broad consensus about environmental protection (Clarke et al., 2009). Although often still regarded as such (Green-Pedersen, 2019, p. 115), it is – due to both new research findings and political developments – increasingly perceived as a conflictual issue on which parties (and other actors) take different positions. Sometimes these positions diverge not only in their preferred public policies, but also in respect of the values and fundamental scientific beliefs that underpin them (Dunlap et al., 2016; Fielding et al., 2012). The emergence of Green parties and the Radical Right have underlined the potential for conflicting positions on the environment (Dickson & Hobolt, 2024; van Haute, 2016)(Gougou & Persico, 2026). Studies focusing on these questions have suggested that these issue characteristics depend on context, and that both position and salience are relevant (Carter & Little, 2021; Lüth & Schaffer, 2022).

A second long-running debate concerns how the environment relates to other, established, political conflicts: is it independent of them (Dalton, 2019) or has it been incorporated into the left-right conflict (Kitschelt, 1988; Neumayer, 2004; Rohrschneider, 1993)? While it is plausible that the political incorporation of environmental issues is complicated by its cross-cutting nature and associated embedded interests on the left and the right (Mildenberger, 2020), the ‘incorporation’ perspective has arguably been predominant in the study of the party politics of the environment (Green-Pedersen, 2019, p. 116).

This study builds on these debates, focusing initially on two questions. The first concerns the *politicisation* of the environment. We know a good deal about the politicisation of the environment among the public (European Commission, 2025; Fisher et al., 2022; Kenny, 2021), but we know less about the politicisation of the environment among parties. In party systems, issue politicisation is frequently conceptualised as involving salience (the priority accorded to the issue) and polarisation

(how widely actors' positions are dispersed) (Hutter & Kriesi, 2022, p. 343; Zürn, 2019).¹ Some research on environmental politics has sought to account for both the 'valence' and positional perspectives on the environment, highlighting that these characteristics differ from context to context, and within countries over time (Carter & Little, 2021; Lüth & Schaffer, 2022; Paterson et al., 2024). Some researchers have formulated strong expectations that the environment will become more salient over time (Derndorfer et al., 2022, pp. 89–91).

The second question is about whether party politics has become '*greener*' by giving greater priority to the environment and adopting more pro-environmental positions over time. While the emergence of the environment onto political agendas suggests 'greening', many parties have struggled to integrate environmental concerns with their existing policy commitments (Carter, 2013; McDaniel, 2025), even as the urgency of addressing environmental problems like climate change has become increasingly clear.

Despite a growing wealth of data and analysis on parties' environmental policy preferences – both salience and positions – we lack a rounded picture of how politicisation of the environment and the greening of party politics varies across countries and over time. We therefore provide a broad cross-national analysis of the politics of the environment at the level of the party system and party-level mechanisms that draws on three important sources of comparative data on parties' environmental policy preferences, and that uses data on both the salience of the environment to parties and their positions on the environment. This is a contribution to the existing literature which – for good reasons – has tended to focus on the party level, on one source of data per study, and with some exceptions (e.g., Abou-Chadi, 2016; Carter, 2013; Carter & Little, 2021; Lüth & Schaffer, 2022) on either salience or position.

Explaining parties' environmental preferences

The mechanisms that produce party system-level structures of party competition on any issue originate in individual parties' engagement with that issue, and therefore the incentives and constraints that shape their engagement. We review explanations associated with vote- and office-seeking incentives, and with constraints associated with party ideology, before drawing on these mechanisms to develop theoretically two explanations for variation in the politicisation of the environment and the 'greening' of party politics.

¹ Hutter and Kriesi (2022, p. 343) argue that the range of actors involved in political debate on an issue can also be an indicator of politicisation. However, as our focus is on parties and party systems, we exclude this aspect of politicisation.

Vote-seeking

Vote-seeking is a central party goal and therefore vote-seeking explanations for variation in parties' environmental preferences have received most attention. On the demand side, *public attitudes* are a powerful explanation for environmental public policy (Anderson et al., 2017; Bakaki et al., 2020; Bromley-Trujillo & Poe, 2020) and one of the mechanisms through which this explanation operates is by shaping parties' preferences. Parties have good reason to listen to the public, as vote choice is influenced by environmental attitudes (Kenny & Langsæther, 2023; Puleo et al., 2025). Spoon et al. (2014) and Schwörer (2024) show that the public salience of the environment influences parties' preferences; there is also evidence in case studies for parties responding to public attitudes on the environment (Carter & Jacobs, 2014; Ladrech & Little, 2019), and this relationship appears to hold even for parties of the Radical Right (Schwörer & Fernández-García, 2024).

On the supply side, research on environmental preferences has focused on competition from Green parties, the Radical Right, and competition among mainstream parties. It is plausible that *Green party* success should trigger accommodative responses among other parties (increased salience, greener positions), particularly their proximate electoral rivals (Meguid, 2008). Green parties are issue-owners on the environment (Abou-Chadi, 2016), and typically have the strongest (highest salience and greenest positions) and most homogenous pro-environmental policy preferences among party families (BirdLife Europe et al., 2024; Carter, 2013; Langsæther, 2023, p. 255). However, evidence of their role in setting the agenda is mixed. While some studies have observed accommodative responses to Green party success (Ladrech & Little, 2019; Spoon et al., 2014), most comparative studies have found that parties were more likely to adopt greener preferences when Green parties were weak, and to de-emphasise the environment when Green parties were stronger (Abou-Chadi, 2016; Carter, 2013, p. 86; Green-Pedersen, 2019, p. 122); that Green success has no detectable effect once public attitudes were accounted for (Schwörer, 2024); or that they have no effect on positions once systemic salience is accounted for (Abou-Chadi et al., 2020).

On the other side of the 'new politics' conflict (Kitschelt 1994), the impact of the *Radical Right* on environmental public policy has been investigated (Berker, 2026; Ćetković & Hagemann, 2020; Jahn, 2021, 2022; B. Lockwood & Lockwood, 2022). These parties' tendency to oppose environmental policies is well-documented (Dickson & Hobolt, 2024; Forchtner, 2019; Grzymala-Busse et al., 2025; Hess & Renner, 2019; M. Lockwood, 2018; Schaller & Carius, 2019), but how they influence other parties' environmental preferences has received little systematic attention. As with Green parties, accommodative responses are theoretically plausible: parties, especially those competing most directly with the Radical Right, should respond by moving their positions in a less green direction -

similar to how Radical Right parties have pushed the immigration policy positions of mainstream parties rightwards (Abou-Chadi & Krause, 2020; van Spanje, 2010).²

Considering their size and agenda-setting potential, it is perhaps surprising that the influence of *mainstream parties* on party competition on the environment has not received more attention. Carter (2013) observed that *mainstream parties* keep tabs on one another's environmental preferences and are central to shaping the agenda (see also Green-Pedersen, 2019). Dynamics of competition among mainstream parties are illustrated in the cases of the UK and Denmark in the late 2000s, when the mainstream centre-right parties moved to seize ownership of the environment, triggering intense competition with the centre-left on the issue (Carter & Jacobs, 2014; Ladrech & Little, 2019). Other cases show that a lack of mainstream competition on the environment can lead to low salience (Carter & Little, 2021; Ladrech & Little, 2019; Marcinkiewicz & Tosun, 2015). These observations on mainstream party influence are closely related to findings on how the party system agenda influences party preferences on the environment (Abou-Chadi et al., 2020; Green-Pedersen, 2019; Schwörer & Fernández-García, 2024). Thus, while their influence is less-studied than Green parties, they are plausibly important supply-side actors when it comes to shaping vote-seeking incentives to engage with the environment.

Office-seeking

Office-seeking considerations – beyond garnering votes and electoral office – also influence parties' environmental preferences, as parties aim to build and maintain coalitions - and destabilise alternative coalitions - by politicising the environment. These incentives are difficult to measure and therefore difficult to capture in a regression framework (Green-Pedersen, 2019, p. 80). One specific problem is that parties' responses to potential coalition partners can be observationally equivalent to those of parties accommodating a successful electoral rival, and so office-seeking responses can be difficult to distinguish from vote-seeking responses.

Scholarly attention to the role of coalitional considerations in shaping parties' environmental preferences has focused on the relationship between Green parties and mainstream parties, especially Social Democratic parties. Green-Pedersen (2019, pp. 117–118, f.125) finds that that coalition considerations drive Social Democratic parties' attention to the environment: they pay more attention to the environment if they need to coalesce with Green or Social Liberal parties, but

² Potential responses in respect of salience are more ambiguous for the radical right than for the Greens. They may raise the salience of the issue while taking anti-environmental policy positions, or they may pay little or no attention to the issue. They differ from the Greens too in their lack of issue-ownership on the environment, which should reduce their incentive to engage with the issue. However, they may use it as a wedge issue (Dickson & Hobolt, 2024).

this incentive is weaker if they need to coalesce with Christian Democratic parties. Other evidence of mainstream parties adapting their environmental preferences due to (the prospect of) coalition with Green or environmental parties – and the absence of adaptation when those incentives are not present – also comes from case studies (Carter & Little, 2021; Ladrech & Little, 2019; Lees, 2000). The increasing success of the Radical Right and its growing focus on opposition to environmental policies (Dickson & Hobolt, 2024) has brought the shaping of the mainstream right’s environmental preferences by coalitional considerations to the fore, but as with vote-seeking responses, we have little systematic analysis of how parties’ environmental preferences adapt to the prospect of coalition with the Radical Right (see Bocquillon, 2024).

Non-environmental policy preferences

Parties follow vote- and office-seeking incentives but, beyond these incentives, it is also plausible that their engagement with the environment is shaped by commitments to other, non-environmental ideas. Maintaining consistency with these commitments might foreclose certain environmental policy options. For example, economically liberal parties might oppose new or increased environmental taxes or more stringent regulations on economic activity. On the other hand, environmental policies may provide opportunities to advance non-environmental preferences, e.g., increased regulation or taxes, more economic growth via the development of green industry, or greater regional autonomy (Enguer 2014; McEwen and Bomberg 2014) .

There is considerable evidence that the environment is a difficult issue for both left and right to integrate with their existing policy platforms, as these non-environmental commitments constrain their responses (Dalton, 2009; McDaniel, 2025; Mildenerger, 2020). We can see connections between left-right positions and environmental policy preferences in cross-national studies (Carter, 2013; Farstad, 2018; Grzymala-Busse et al., 2025; Schwörer, 2024) and in case studies (Båtstrand, 2014; Carter & Little, 2021; Forchtner & Kølvrå, 2015; Huber et al., 2021; Ladrech & Little, 2019; Marcinkiewicz & Tosun, 2015). There is less clear evidence demonstrating the independent effect of ideology, controlling for other (especially vote-seeking) incentives, although Lockwood (2018) points to the explanatory power of right-wing populist parties’ ideologies – and not only vote-seeking incentives – in explaining their anti-climate policy positions. This association between ideology and policy preferences is also evident in some studies of the relationship between parties’ left-right positions and governments’ environmental policies (Fankhauser et al., 2016; Schulze, 2021; Tobin, 2017; Wen et al., 2016).

Summary

Overall, research on parties' environmental preferences has accumulated into a substantial body of theoretically-informed empirical knowledge. Yet gaps and inconsistencies remain. Vote-seeking mechanisms occupy much of the research agenda, with public attitudes appearing to play an important role. On the supply side, it is theoretically plausible that parties with outlying preferences shape preferences, but the evidence is mixed on Green parties, while the influence of the Radical Right is under-explored. Office-seeking incentives have been examined in a few comparative case studies, but not through broader comparative studies. Multiple accounts suggest a relationship between ideology and environmental preferences, but have not established the independent effect of parties' non-environmental policy preferences. We revisit some of these lines of inquiry, but we also give attention to two other arguments, both of which have received little attention to date.

Theory

Hypothesis 1: Problem pressure, greening and politicisation

The first argument concerns the role of *environmental problem-pressure* in shaping parties' environmental policy preferences. Policy agenda-setting research has shown that policy problems play a predominant role in allocating policymakers' attention (e.g., Jones & Baumgartner, 2005, ch.3), and this applies to parties too (Kristensen et al., 2023). The environment is an issue for which long-term policy problems and focusing events are important, indicators of which vary in severity over time and between countries (Block et al., 2024). And problem perceptions are arguably key aspects of parties' policy preferences (Pollex & Berker, 2022).

It is plausible that to meet societal expectations concerning the problem-solving role of politics, and thus to retain and gain voter support, parties' preferences should be responsive to policy problems (Kristensen et al., 2023). In this way, the influence of problem pressure operates mainly through vote-seeking mechanisms.³ Environmental problems may elicit attention to the environment, 'problem-solving' policy proposals and greener positions from parties, but it may also lead parties to propose diverse diagnoses of the problem and diverse solutions, and thus to disagree (and polarise) on the environment.

³ However, it is also conceivable that solving problems is driven by the role-conceptions of parties and their leaders as problem-solvers.

Previous studies have focused on how problems influence party competition on economic issues (Kristensen et al., 2023, p. 2860), and empirical research on the role of environmental problems in shaping parties' environmental preferences is limited. Case studies focusing on parties and environmental policy take account of the 'problem stream' within an agenda-setting approach (Carter & Jacobs, 2014; Green-Pedersen & Wolfe, 2009). Green-Pedersen (2019, p. 123) examines this relationship in comparative perspective across seven party systems, providing the broadest study to date, and showing that the greater are environmental problems, the more attention parties pay to the environment. Kristensen et al. (2023) show that problems shape parties' issue-focus, but they do not report results on the environment separately.

Overall, environmental policy problems are plausible causes of increased politicisation of the environment (both increased salience and polarisation) and the 'greening' of party politics (increased salience and greener positions). However, analysis of their effects remains limited, especially with respect to position(s) and polarization.

Hypothesis 2: Historic conflict structures and politicisation

The second argument that we test intersects with the long-standing question of how the environment relates to the broader political terrain, including to what extent and under what conditions it is independent of (Dalton, 2019) or incorporated into established conflict structures (Kitschelt, 1988; Neumayer, 2004; Rohrschneider, 1993). We use Green-Pedersen and Little's (2024) argument concerning the relationship between historic conflict structures and 'new politics' issues to investigate this question, and specifically to interrogate the idea that the *politicisation* of the environment depends on the durable incentives provided by the long-term strength (or weakness) of the left-right conflict. Green-Pedersen and Little (2024) argue that the strength of the primary conflict structure – the left-right meta-conflict in many systems – reduces uncertainty associated with the politicisation of 'new' issues, including the environment. They focus on issue-attention as an indicator of politicisation, and they find support for the argument in the 'negative case' of Ireland, where the weak politicisation of 'new politics' issues is traceable to the weakly-defined conflict at the core of the party system, despite other conditions for the politicisation of the environment being in place.

Green-Pedersen and Little (2024) highlight the reduction of *vote-seeking uncertainty* – uncertainty about how voters will respond to the politicisation of the new issue – as one mechanism through which a strong historic conflict can shape the politicisation of new politics issues (M2.1). Where there

is strong left-right conflict, parties can be more certain about how *their voters* will respond, and therefore will engage with the issue, especially where it can give them a vote-seeking advantage. This engagement will be reflected in party competition on new issues, particularly at the core of the party system.

They also argue that a strong historic conflict can reduce uncertainty about how potential *coalition partners* will respond to the politicisation of new politics issues (*office-seeking uncertainty*) (M2.2). For example, Social Democratic parties will anticipate that adopting pro-environmental preferences reinforces their chances of being Green parties' go-to coalition partner (Green-Pedersen, 2019, pp. 114–134). Similarly, the environmental preferences of Conservative parties might be shaped by potential coalition with the Radical Right.

To these two mechanisms we add a third: that a stronger overall conflict structure will be associated with stronger (more outlying, or more important) *non-environmental policy preferences*, which therefore more strongly shape parties' environmental policy preferences (M2.3). We expect that the stronger the left-right conflict, the more 'new' issues like the environment are influenced by left-right preferences. This tendency may be rooted in anticipation of how actors other than voters and potential coalition partners will respond (e.g., how intra-party actors will respond).

Overall, when the primary conflict is strong, new issues are more likely to be incorporated into it in a stable and predictable way, and this process can reduce uncertainty associated with their politicisation for parties. In this scenario, parties have greater certainty about how key actors (voters, coalition partners, intra-party actors) will respond to the politicisation of new issues. The structural character of historic conflict makes for a durable set of incentives across multiple elections. For instance, it may lead to the environment being regarded as a 'left' issue, or particular positions on the environment being associated with the left and others with the right. Focusing on the environment, we aim to test this argument in a cross-national perspective, and by examining polarisation as well as issue-salience.

Hypotheses

In summary, we set out to test two main hypotheses:

H1: The more acute are environmental problems, the more politicised is the environment and the 'greener' are party preferences.

The main logic underpinning this hypothesis is vote-seeking.

H2: The stronger the historic left-right conflict, the more politicised is the environment.

Building on Green-Pedersen and Little's (2024) argument, we have identified three mechanisms through which we expect historic conflict (H2) to influence the politicisation of the environment. We examine whether the strength of the historic left-right conflict influences the politicisation of the environment, and then whether these mechanisms mediate any such effect.

Our third and fourth hypotheses are grounded in some of the main explanations identified in our literature review, and specifically vote-seeking explanations. First, we expect public attitudes to influence both the politicisation of the environment and the 'green-ness' of parties' environmental preferences.

H3. The more pro-environmental are public attitudes, the more politicised is the environment and the 'greener' are party preferences.

We also investigate the role of niche parties in shaping (other) parties' preferences. The idea that a stronger Green party would compel other parties to accommodate their environmental preferences is theoretically compelling, but is at odds with much of the available research evidence.

H4a. The stronger are Green parties, the more politicised is the environment and the 'greener' are party preferences.

And the role of Radical Right parties in influencing other parties' environmental preferences has not received much attention, even as it is increasingly relevant.

H4b. The stronger are Radical Right parties, the more politicised is the environment (polarisation only) and the less green are party preferences (position only).

Expectations associated with each explanation in respect of politicisation (salience and polarisation) and greenness (salience and position) are summarised in Table 1.

Table 1. Expectations regarding party competition on the environment

<i>Indicator type (outcome concept(s)):</i>	<u>Salience</u> (politicisation and greening)	<u>Position</u> (greening)	<u>Polarisation</u> (politicisation)
Development over time (1990 - early 2020s)	+	+	+
H1. Environmental policy problems	+	+	+
H2. Historic left-right conflict	+	No expectation	+
M2.1. Vote-seeking incentives	+	+*	+
M2.2. Coalition- building incentives	+ w/Green	+ w/Green – w/RR*	+
M2.3. Ideology (left)	+	+*	No expectation
H3. Public attitudes on environment	+	+	+
H4a. Niche party competition (Greens)	+	+	+
H4b. Niche party competition (RR)	No expectation	–	+
Each cell first indicates the direction of the hypothesised relationship: +/– = more/less polarisation on the environment; greater/less salience of the environment; or greener/less green positions on the environment. * We do not have theoretical reasons to expect historic conflict to influence positions on the environment, so we test these factors as independent variables in their own right rather than as mechanisms that mediate the effects of historic conflict.			

Research design and methods

The analysis proceeds in three steps. First, we ask whether politicisation and ‘greening’ have increased over time, drawing on three sources of comparative data. Second, we examine the relationship between structures of competition on the environment at the system level (specifically, the politicisation of the environment and the greening of party politics) and three of our key independent variables at the system level (environmental problems, historic conflict, public attitudes). Third, we analyse the role of these factors in influencing party preferences at the party level. The party-level analyses also allow for an examination of H4a and H4b (niche parties), and for investigation of the mechanisms that we suggest mediate the effect of historic conflict on the politicisation of the environment.

Outcome variables

Our dependent variables measure politicisation of the environment and the ‘greening’ of party systems, drawing on indicators of the salience of the environment to parties and parties’ positions on the environment. *Salience* is both an indicator of the politicisation of the environment *and* the ‘greening’ of party politics. We use the *per501* variable from the Manifesto Project, which measures the percentage of manifesto content (quasi-sentences) that focuses on ‘General policies in favour of protecting the environment, fighting climate change, and other “green” policies.’ (Lehmann et al., 2025, p. 18). Compared to the main alternative measure of salience in expert surveys, *per501* has the advantage of being a direct measure of salience (not mediated by expert opinion). It allows for more variation over time than expert surveys, as well as having greater empirical coverage.⁴

For the salience of the environment in party manifestos and other CMP-based data on positions (and therefore polarisation), we use data from 1990 to 2025.⁵ This restricts the scope of our analyses, but also affords some advantages: for the descriptive analyses, it allows us to include Central and Eastern European countries (more or less) throughout the data set, without major compositional changes over time in respect of countries ([reference to coverage table/heatmap in appendix here](#)); for the explanatory analyses, it allows us to investigate Hypothesis 2, which is operationalised using a 30-year average polarisation on the right-left dimension.

⁴ Expert survey data on salience provides data points from 2003, 2010, 2019 and 2024 (Benoit & Laver, 2006; Rovny et al., 2025).

⁵ However, the years 2023, 2024 and 2025 include only four elections (Greece 2023, Spain 2023, UK 2024 and Germany 2025); we enter caveats about observations from these years in the descriptive analyses.

Parties' *positions* on the environment have been estimated in a number of ways. We draw on three sources of positional data.

- 1) *Manifesto content*. We use data from the CMP, and following Abou-Chadi (2016)⁶ we operationalise parties' positions on the environment as a ratio, which indicates the balance between content categorised as *per501* (Environmental Protection) and *per410* (Economic Growth: Positive):

$$Position = (per501 - per410)/(per501 + per410) \quad (1)$$

- 2) *Expert survey data*. We use the 'environmental protection vs economic growth' variable available in expert surveys from 2003, 2010, 2014, 2019 and 2024, which indicates the extent to which the party supports environmental protection, 'even at the cost of economic growth' and vice-versa (Benoit & Laver, 2006; Rovny et al., 2025).⁷
- 3) *EU&I VAA data*. We use two environmental policy items that have been coded for parties across four rounds of the EU&I Voter Advice Application (2009-2024) using evidence-based coding by parties and experts (Cicchi et al., forthcoming; Ferreira da Silva et al., 2023). Parties are positioned on a five-point Likert scale (completely agree to completely disagree) on these items. The first item relates to public transport and green taxes ('The promotion of public transport should be fostered through green taxes (e.g. road taxing)') and the second is on renewable energy ('Renewable sources of energy (e.g. solar or wind energy) should be supported even if this means higher energy costs'). We calculate parties' mean position across these two items.

We use all three indicators because each are potentially valid measures of parties' positions, and because each has limitations: both EU&I (2009-2024) and expert survey items on environmental positions (2003-2024) have more limited over-time coverage than manifesto data; both the expert survey and manifesto-based measures counterpose the environment to economic growth, which

⁶ See the Appendix for further discussion of this operationalisation and available alternatives calculated from CMP data.

⁷ Benoit and Laver's (2006) data is transposed to the same scale as the CHES item. The wording of these items differs slightly: the 'green' end of the CHES scale indicates that the party 'Strongly supports environmental protection even at the cost of economic growth', while Benoit and Laver's (2006) indicates 'Supports protection of the environment, even at the cost of economic growth.' The pro-growth end of the scale indicates 'Strongly supports economic growth even at the cost of environmental protection' in CHES, and 'Supports economic growth, even at the cost of damage to the environment' for Benoit and Laver (2006).

some argue is outdated; the EU&I items cover only a subset of environmental issues; and the expert survey measure is indirect (mediated by experts) and varies little over time.

We measure position at the system level by weighting parties' positions by seat share, i.e., by calculating the system's 'centre of gravity' on the environment (Gross and Sigelman, 1984). We also use the positional measures described above to estimate *polarisation* on the environment at the country level, following Dalton (2008):

$$Polarisation = \sqrt{\sum_{i=1}^n \left(\frac{(pos_i - wmean)}{100} \right)^2 \cdot S_i} \quad (2)$$

where Polarisation is the environmental polarisation index ranging from 0 to 10, pos_i is a party's environmental position, $wmean$ is the ideological centre of gravity defined as the mean environmental position weighted by the parties' seat share, and S is a party's seat share.

At the party level, we estimate 'polarisation' as the extent to which a party contributes to system-level polarisation on the environment, i.e., its distance from the system's centre-of-gravity (Gross & Sigelman, 1984; for a similar measure, see Kriesi et al., 2026).

$$Party\ polarisation = pos_i - \sum_{i=1}^n \left(\frac{S_i}{T} \cdot pos_i \right) \quad (3)$$

In sum, we have three system-level dependent variables (salience, position, and polarisation on the environment) and three party-level dependent variables (salience, position and the party's contribution to polarisation on the environment). We use one data source for salience (CMP's per501), and three for position and polarisation (positions derived from CMP variables, an expert survey item, and two VAA items). At the system-level, these data allow us to assess politicisation of the environment (indicated by salience and polarisation) and the 'greening' of party systems (indicated by salience and position), while party-level data allow us to study how parties' responses contribute to these system-level structures of party competition on the environment, and what shapes those responses.

Independent variables

Table X describes our main independent variables. All are lagged by one election. We use greenhouse gas emissions as an indicator of *environmental problems* (H1). There is of course a wide range of environmental problems, not only GHG emissions. However, GHG emissions are correlated with measures of man. Countries with higher GHG emissions tend to experience greater air-pollution burdens (such as ozone exposure, lead exposure, and household air pollution), generate more municipal waste, and exhibit weaker ecosystem resilience, indicating that emissions broadly track underlying environmental stress. Further discussion of these issues appears in the appendix.

The *historic conflict structure* (H2) is a system-level attribute. Green-Pedersen and Little's (2024) argument applies, in principle, to any primary conflict, but as their empirical focus was on Ireland and how it compared to West European countries, they focus on the left-right meta-conflict. While we acknowledge that the primary conflict can differ in other systems, notably in Central and Eastern Europe (Mölder, 2016), we likewise assume that the left-right conflict is the primary meta-conflict. We measure the strength of the primary conflict as mean polarisation on the left-right dimension across previous decades. In order to draw on a measure of left-right polarisation that can take account of polarisation in decades before our timepoints of interest (election year for CMP data, survey year for CHES data), we make use of polarisation on the CMP's *rile* measure, calculated using Dalton's (2008) Polarisation Index.⁸ As we are interested in durable incentives set by long-term systemic structures, we measure this mean polarisation over 30 years (with the mean for each legislative period weighted by years). For our analysis of environmental preferences, there are other considerations associated with this choice. The CMP *rile* indicator does not encompass the CMP environment code (*per501*), nor *per410* which we use in our measure of environmental policy position. Therefore, in analyses of CMP data, there is no empirical overlap between our polarisation variable (independent variable) and measures of environmental preferences (dependent variables).

[Could add a sentence here on face validity of this measure, i.e., that countries we consider polarised are polarised, with a reference to a panel of graphs in the appendix] [Add in note on why we use HC from CMP in models with DVs from other sources]

We identified three factors that might mediate the effect of historic conflict on the politicisation of the environment, and which may also influence parties' environmental policy positions. For the vote-seeking mechanism, we examine how *party competition on the environment* (M2.1), in the form of parties' politicisation of the environment, mediates the effect of historic conflict on politicisation, as parties are compelled to respond to the 'party system agenda' (Green-Pedersen & Mortensen,

⁸ https://manifesto-project.wzb.eu/download/do_files/mpelds_documentation.pdf

2010).⁹ We measure this as systemic (weighted) salience or polarisation, lagged (see section above on dependent variables). [Need to explain here how we use these two options.] We noted above that measuring *coalition-building incentives* (M2.2) is difficult. We suggest that niche party strength presents particularly strong incentives in more proportional systems (and less in more disproportional systems). Therefore, we interact a measure of disproportionality (the Gallagher Index) with the seat strength of Green or Radical Right parties. When these parties are strong in a more-proportional system, other parties will have incentives to tailor their preferences so that coalition with them is more feasible. However, we acknowledge that these interaction effects may not only reflect office-seeking incentives (M2.2); they may also reflect vote-seeking incentives in those systems (H4). To estimate M2.3 (how non-environmental preferences shape parties' environmental preferences), we measure ideology at the party level as the party's left-right position, using CMP's *rile* measure.¹⁰

Finally, we measure *public attitudes on the environment* (H3) by using Eurobarometer's 'Most Important Issue' item, and thus as the public salience of the environment and we measure electoral competition from *niche parties* (H4) as the strength of Green and Radical Right parties (separately). We also control for GDP per capita using data from the OECD.

The scope of country and temporal coverage is broadly consistent with the set of European democracies included in our CMP, the Expert Surveys, and the Voting Advice Voting Advice Application data sources: between 25 and 30 European countries from the early 2000s to the early 2020s. There are only minor differences in country coverage across these datasets given that Expert Survey data additionally include Malta, while CMP includes Switzerland, Norway, Cyprus and Luxembourg, which are not covered in the VAA data. The independent variables draw on external datasets with wider geographic and temporal frames (for instance, GHG emissions available globally from 1970–2023 and GDP per capita from 1960–2023), but when merged to the party–position datasets their composition aligns closely with the European sample used in the models.

⁹ While Green-Pedersen (2019) and Green-Pedersen and Little (2024) stress the responses of large mainstream parties, we see few reasons why this should not apply to other parties. This is reflected in the operationalisation of the relevant variables, below.

¹⁰ We do not expect historic conflict to be associated with position on the environment, so for the positions models (both system- and party-level), we investigate systemic environmental salience, coalition-building incentives and left-right position as independent variables in their own right, rather than as mechanisms that link historic conflict to parties' positions on the environment (see theoretical expectations table above).

Method

The analysis proceeds in two main steps. First, we test H1, H2 and H3 at system level. Second, we test them at party level, and we examine whether M2.1, M2.2 and M2.3 mediate the effects of historic conflict (or have effects that are independent of historic conflict).

Given the structure of the data – political parties nested in elections in different countries – we rely on hierarchical or multi-level models. In the system-level analysis, random intercepts at the election and country levels indicate that the baseline level of our outcome variables systematically varies across these grouping factors. Moreover, at the party level, random intercepts capture the interdependence among political parties from the same electoral contexts. As such, given our assumption that some countries may naturally have higher or lower environmental polarization due to unique country or election-specific factors, multi-level models allow us to model this heterogeneity to correctly estimate standard errors (Bell & Jones, 2015).

To estimate the effect of our main outcome variables on the environment (salience, position, and polarisation), we rely on multilevel linear models with random intercepts for countries and election years. Our specifications include lagged and standardised country-level predictors (greenhouse gas emissions, public opinion, GDP per capita, systemic environmental salience, and the ideological centre of gravity) alongside party-level characteristics such as the seat shares of Green and Radical Right parties and the disproportionality of the electoral system. To capture the influence of niche parties (office-seeking, i.e., M2.2 or vote-seeking i.e., H4), we additionally interact the lagged seat shares of Radical Right and Green parties with the lagged disproportionality (Gallagher Index). All explanatory variables are mean-centred and standardised to improve stability and interpretability, and prevent variables with larger scales from dominating the model. Random intercepts for countries and years account for unobserved heterogeneity in baseline environmental positions across contexts and over time.

Our first set of models are at the system level and test three system-level hypotheses: H1 (GHG emissions), H2 (historic conflict) and H3 (public opinion).¹¹ As the dependent variables are weighted by seat share, these system-level models are key tests of our hypotheses. However, they also involve relatively small numbers of observations (75 country-years for CMP data, 60 for expert surveys, and 52 for VAA data).

¹¹ Testing H3 (niche parties) at system level would involve having niche parties on both on the independent variable and dependent variable side of the equation, potentially biasing the estimates.

The party level models allow us to examine parties' responses to the hypothesised causal factors. At the party level, the unit of analysis is a country year¹² observation and we rely on multilevel linear regression with random intercepts for years and countries, to account for the clustering of parties within electoral contexts and within national systems.

We then estimate each of our expectations in a stepwise manner. First, we estimate the effects of our main explanatory variables (lagged environmental problems, the strength of historic conflict, and public salience of the environment, while controlling for economic growth) in a baseline specification that includes *all parties*. This baseline model captures the direct association between these system-level conditions and parties' environmental preferences. As it includes all parties, we regard it as the key test of H1, H2, and H3.

Then, we specify a model that tests H4a and H4b (on niche parties) by including variables that measure the strength of Green parties and radical right parties, to test the effect of supply-side vote-seeking pressures. We *exclude both niche party families* from the sample so as not to bias the results.

Then we test each of the variables associated with the 'historic conflict' mechanisms in turn. First, we add a lagged variable for systemic politicisation of the environment,¹³ in order to test whether general party-system competition on the environment mediates the effects of historic conflict (M2.1). Second, we examine whether the influence of niche parties is conditional on the proportionality of the system (M2.2): first we do this for Green parties, while excluding Greens from the model, and then for the Radical Right, while excluding them from the model. Third, to test whether the effect of historic conflict is mediated by prior ideological constraints we then test parties' left-right positions.

Finally, we include two further models. The "Full (no interactions)" model includes system-level and party-level variables, as well as the politicisation and left-right variables, but we omit interactions with the Gallagher Index. The final specification ("Final") incorporates all variables that showed significance in one of the previous models. Here, because of the limited number of observations relative to the number of covariates and interactions, we include only the interaction terms between niche-party strength and proportionality that displayed a substantively meaningful and statistically significant marginal effect in previous models.

Robustness:

- For robustness, we also run a version of the full model that excludes the niche variables (and therefore includes the niche parties) (see appendix).

¹² Elections for CMP data, survey-years for expert survey data, and European Parliament election-years for VAA data.

¹³ Lagged environmental salience with the models of positional dependent variables (position or polarisation) and lagged polarisation in the models of environmental salience, in order to avoid multi-collinearity.

- We also run a set of models (or perhaps just ‘full’ models) that substitute contemporary rile polarisation for historic conflict (see appendix).

Results

Politicisation and greening over time

Salience is best characterised as cyclical. The rolling average for the salience of the environment in party manifestos shows that salience declines from the early 1990s, rises in the mid- and late-2000s, slumps after 2009 and remains relatively low until it rises again in the late 2010s, and then begins to fall in the early 2020s. There is little evidence that the salience of environmental issues has increased over time, for the level of salience in the early 1990s is comparable to the level of salience in the early 2020s. If anything, salience may have declined over the long term, which would indicate ‘de-greening’ and reduced politicisation.

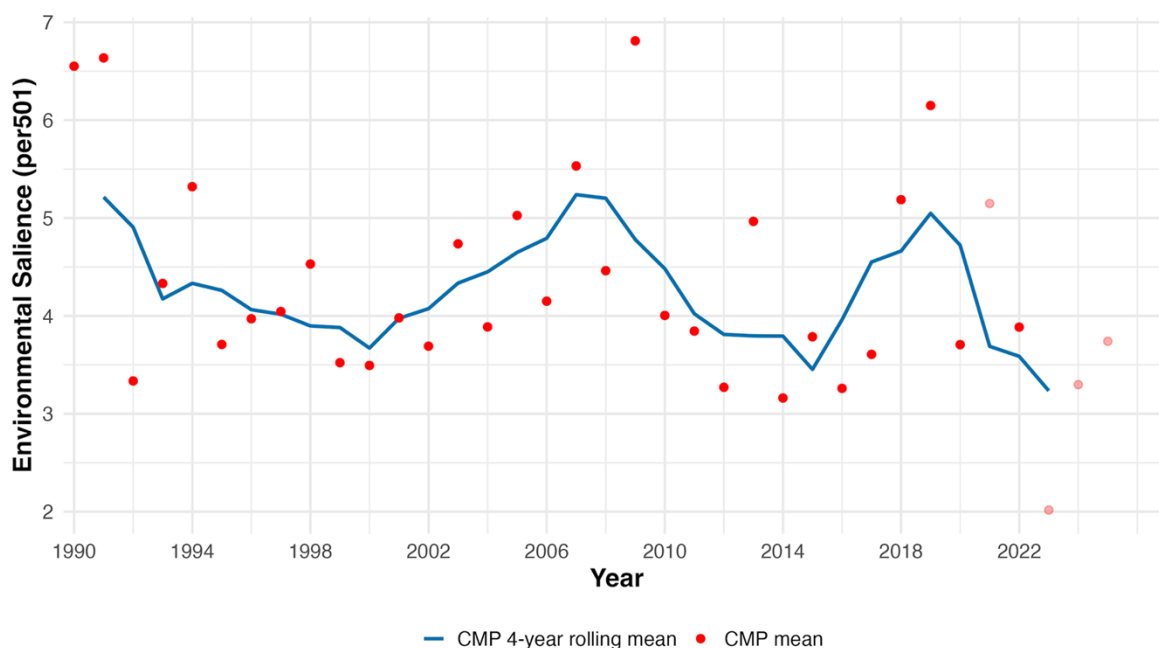


Figure. The salience of the environment to parties over time (per501 environmental protection: mean of country values, which are seat-weighted country means). 2021, 2023, 2024 and 2025 are shaded because they represent only one or two countries each.

The CMP data shows that parties’ environmental *positions* – a second indicator of ‘greening’ – remained steady during the 1990s and 2000s, and became less green through the 2010s before becoming greener at the end of the decade, with peak ‘greenness’ values in 2019 and 2021

(surpassed only by 1990). Expert survey and VAA data suggest a ‘de-greening’ trend – after 2003 (expert survey) and 2009 (VAA) – that is consistent with the CMP trend. In the expert survey data, this trend accelerates in 2024, whereas in the VAA data, the negative trend is arrested in 2024.¹⁴ For all three data sources, the observed range of variation in annual country means is substantial: about one point (5.85 to 4.92) on an eleven-point scale for expert survey data, from 60 to [49?] on a 100-point scale for the VAA data, and from X to Y for CMP data.

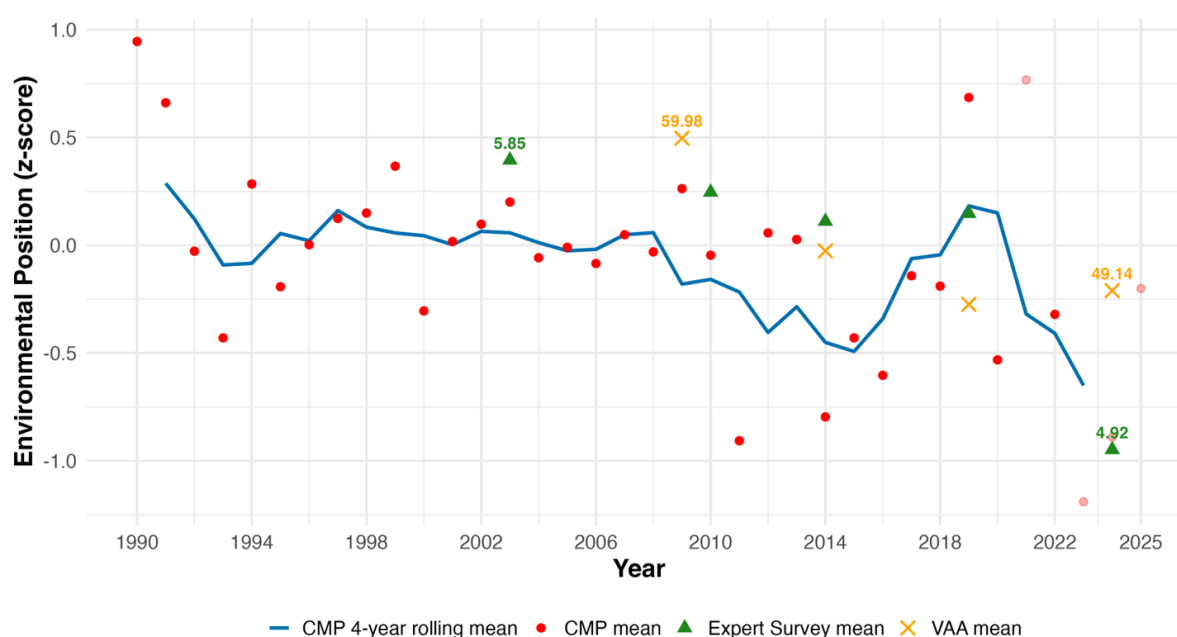


Figure. Parties’ positions on the environment over time (mean of country values, which are seat-weighted country means). CMP data points for 2021, 2023, 2024 and 2025 are shaded because they represent only one or two countries each.

All three data sources suggest rising *polarisation* on the environment over time, with some exceptions: depolarisation in CMP data in the early 1990s and slight depolarisation in expert survey data between 2003 and 2010. The polarisation values for CMP data suggest sustained increases in polarisation in the mid-2000s and then in the late 2010s. There is a decline in the rolling average going into the early 2020s, but it remains for the most part above pre-2015 values.

¹⁴ There are too few country observations in the CMP data for 2023, 2024 and 2025 (only 4 overall) to make a direct comparison with this later timepoint.

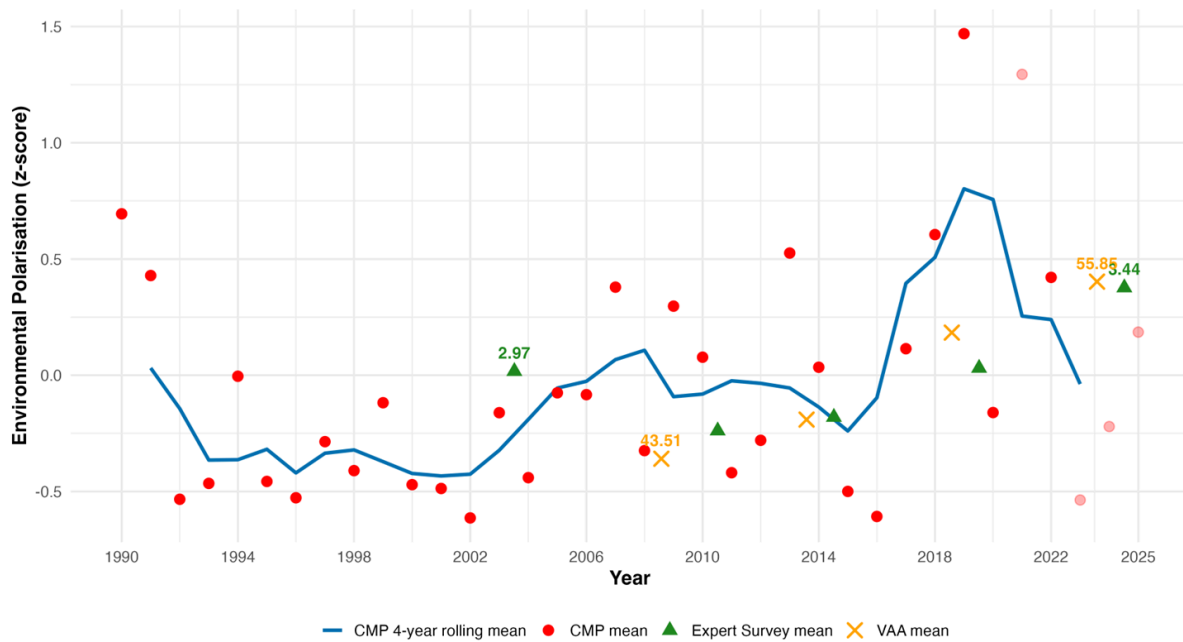


Figure. Polarisation on the environment over time. CMP data points for 2021, 2023, 2024 and 2025 are shaded because they represent only one or two countries each.

Explaining variation in environmental policy preferences

What explains variation in politicisation and ‘greening’? In order to examine the hypotheses developed above, we look at each of the indicators in turn: salience (an indicator of politicisation and greening), position (greening) and polarisation (politicisation). Figures X-Y plot the results of system-level models that test H1, H2 and H3; a party level model that includes all parties and tests H1, H2 and H3 (‘baseline’ models); and a second party level model that tests H4a and H4b and therefore excludes niche (Green and radical right) parties. These ‘main models’ are available in the Appendix, Table X-Y. They are complemented by other models that test mechanisms 2.1, 2.2 and 2.3; H3 (public opinion) without controlling for GDP; and the robustness of the results concerning H4a and H4b by testing the effects of Green and radical right parties individually (rather than together) (Appendix, Tables X-Y).

Figure X shows results from the ‘main models’ of salience (Appendix, Table X). Historic conflict (H2) is associated with positive and statistically significant effects on environmental salience, both at system

level and at party level. It is the only variable that is consistently significant across 'main model' specifications. Although coefficients are positive, as expected, for GHG emissions (H1) and public attitudes (H3) they are not statistically significant. Public attitudes (but not GHG emissions) do, however, become significant at system level when GDP is excluded from the models (Appendix table X, model X), which suggests that it may mediate the effects of GDP on systemic salience.

There is no evidence that Green parties have the expected effect of raising salience; the coefficient is negative (H4a). There is limited evidence that stronger Radical Right parties lead to reduced salience of the environment among other parties ($p < 0.1$), albeit we did not have an expectation about this. These results are robust to excluding only Radical Right or Green parties (rather than excluding both) (Appendix, Table 4, models 1 and 2).

As regards the three proposed mechanisms for historic conflict (models 3-7 in Table 4, Appendix), only ideological position (M2.3) shows consistent and statistically significant effects: parties further to the right (left) give less (more) salience to the environment. However, introducing this variable does not diminish the effect of historic conflict or make it less significant, casting doubt on whether it mediates its effect. Environmental polarisation (M2.1) does not have the expected positive effect on salience, and so does not mediate historic conflict in the expected way. Of the niche parties (M2.2), there is evidence for a negative Radical Right effect (Table A4, models 5 and 7) mainly in contexts with somewhat above-average values of disproportionality (Figure A1, Appendix), contrary to the expectation that niche parties' effects would be strongest in more proportional systems.

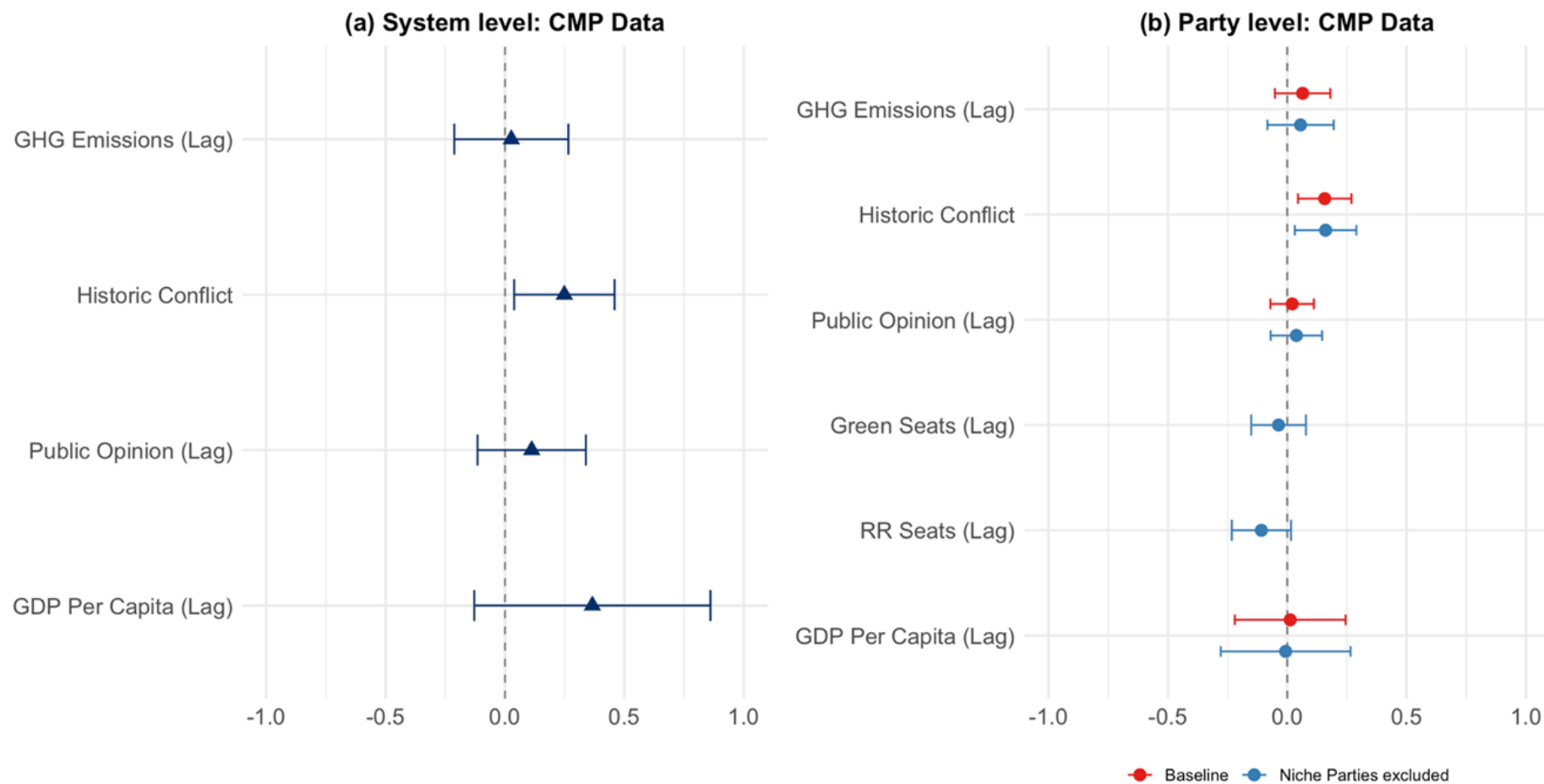


Figure. Environmental salience models (for details, see Table AX, models 1-3).

Turning to variation in parties' positions on the environment, there is contradictory evidence across data sources for the 'greening' effect of environmental problems (H1) on policy positions (Figure X). Analysis of CMP data suggests clear positive effects of environmental problems at system and party level, while the effects on position are negative for expert survey data and VAA data at system level, and these coefficients remain negative (but not significant at $p < 0.05$) at party level. Although we did not have any expectation that historic conflict (H2) would influence position, it is notable that there is (similarly) evidence of substantial positive effects in CMP data at system- and party-level, but no such evidence in the analyses of the other two data sources.

Public opinion (H3) has no clear effects on position across analyses and data sources in Figure X, although in most models the coefficients are positive, as expected. GDP is associated with a clear positive effect on parties' environmental positions at system level, but not at party level. When GDP is omitted from the models, public opinion becomes significant and positive in system-level models of CMP and expert survey positions (Appendix, table X, models X and Y), lending some support to H3.

There is no evidence in these models to support the expected 'greening' effect of Green party strength on parties' environmental positions, and there is also a notably large negative coefficient (albeit not statistically significant) associated with Green party strength in the analysis of VAA data. There is some evidence, however, of one clear niche party effect on environmental positions: in the analysis of expert survey data (panel c in Figure X), Radical Right strength (H4b) has a large negative effect on parties' positions. However, this finding is not reflected in the analysis of CMP or VAA data. These results are robust to excluding only Radical Right or Green parties (rather than excluding both) (Appendix).

As we had no theoretical expectation that historic conflict would influence position, there is little reason to examine potential mechanisms of historic conflict. But further analyses show that, as in the CMP salience models, the party's left-right position (M2.3) has the expected effect (parties further to the right take less green positions: model 6), but shows no sign of diminishing the effect of historic conflict, so we cannot claim that it is mediating its effects.

These further analyses also add to our knowledge of niche party effects. Although there is no net Green party effect, further investigation shows that, in analyses of CMP data, Green party strength in highly proportional systems leads to parties taking greener positions, while in highly disproportional system, it leads to the opposite: less green positions. And in the analyses of the expert survey data, the analyses confirm that the negative effect of Radical Right parties on environmental position is in

evidence across both disproportional (except the most disproportional) and proportional systems.¹⁵

In the VAA data, a negative Green party effect (i.e., that parties become less green when Green parties are stronger) is evident in proportional systems.

¹⁵ The final ('full') expert survey (Appendix, table X, model X) suggests that parties further to the right take greener positions. However, this is in the presence of a high number of controls.

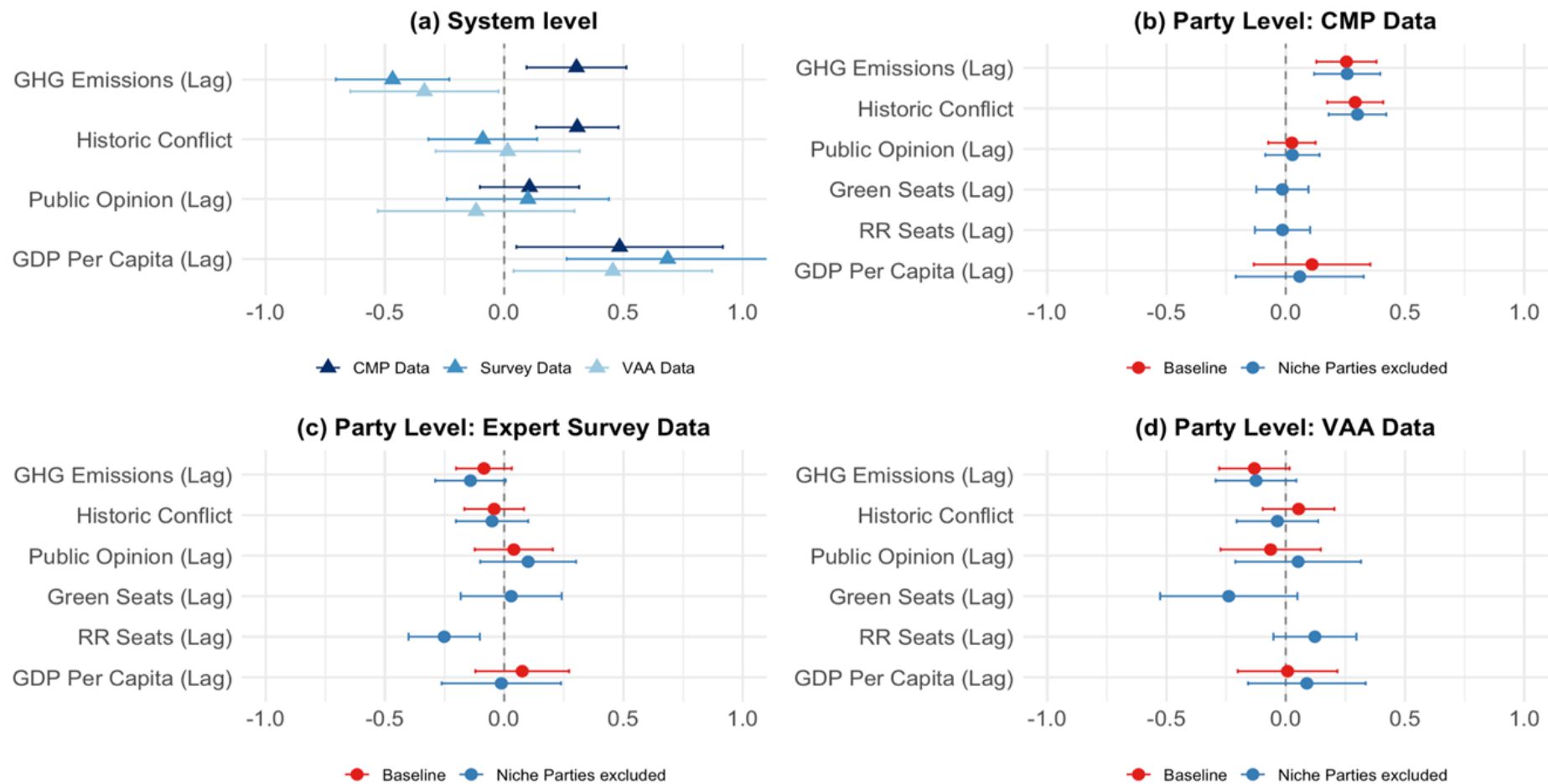


Figure. Environmental position models (see Table AX for details)

In the system-level analyses of polarisation, there is evidence that environmental problems (GHG emissions, H1) are associated with higher polarisation on the environment in CMP data; in the analyses of other data sources, the system-level coefficients for GHG emissions are positive but not statistically significant. The pattern is similar across the main party-level models (Figure X): in almost all cases (with the exception of the baseline VAA model) the coefficients for GHG emissions are positive. They are significant at $p < 0.05$ in the analyses of CMP data and in one of the two main party-level models of expert survey data, lending some support to the idea that greater environmental problems will elicit more outlying positions from political parties.

Similarly, historic conflict (H2) is associated with positive coefficients across nearly all models (again, the exception is the party-level baseline VAA model) and it is consistently significant in CMP models, is significant at $p < 0.1$ in the expert survey baseline party-level model, and becomes significant and positive in the analyses of VAA data when niche parties are excluded.

The coefficients for public opinion (H3) are positive at system level, as expected, and are positive for nearly all party-level models. They are statistically significant at $p < 0.05$ only for VAA models (system-level and baseline party-level), but are also significant at $p < 0.1$ for the system-level CMP analysis. When GDP is omitted from the system-level models, public opinion becomes significant and positive in the analysis of expert survey data, remains significant and positive for VAA data, and remains significant at $p < 0.1$ in CMP data (Appendix, table X, model X).

There is some evidence of niche party effects on polarisation. In expert survey data and VAA data, there is evidence for Radical Right parties (H4b) having a strong positive effect on polarisation on the environment, while the coefficients associated with Green party strength are negative (but not significant). There is no evidence to support H4a or H4b in the CMP data, although there is some evidence of conditional effects (M2.2, described below). These results are robust to excluding only Radical Right or Green parties (rather than excluding both) (Appendix).

As there is some evidence for an historic conflict effect (H2), we further examine the evidence for the mechanisms that we expect to mediate its effect. There is no evidence for systemic politicisation (salience) being associated with parties' contribution to polarisation; nor is there evidence for an effect of ideology (CMP and expert survey analyses; these mechanisms are not tested for VAA data).

With regard to niche parties in different electoral systems, the evidence varies. For Green parties, their effects on parties' contribution to polarisation are not significant across all values of disproportionality, regardless of which data source is used. The expert survey and VAA analyses of Radical Right effects yield similar results: in the analyses of expert survey data, the positive effect of

Radical Right parties on other parties' polarisation becomes statistically significant in less proportional systems (contrary to expectations about office-seeking incentives); in VAA models, the positive effects of Radical Right parties on parties' contribution to polarisation becomes significant in all but the most proportional systems, where their effect is indistinguishable from zero. The opposite is evident in the analysis of CMP data, where the positive effect of the Radical Right on environmental polarisation is particular to more-proportional systems.

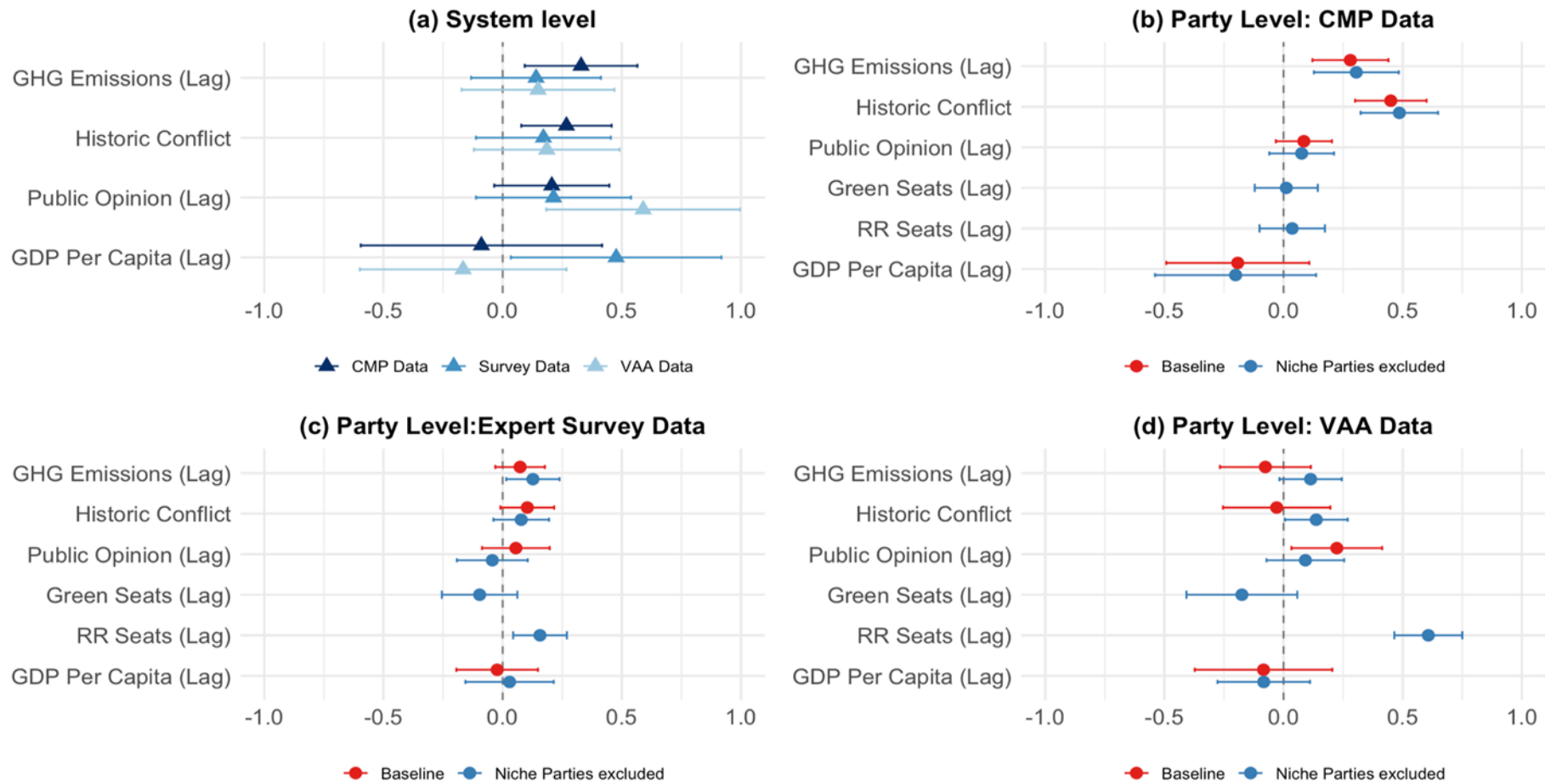


Figure. Environmental polarisation models (see Table AX for details)

Table. Overview of results.

<u>Indicator type</u> (outcome concept(s)):	<u>Salience</u> (politicisation and greening)	<u>Position</u> (greening)	<u>Polarisation</u> (politicisation)
Development over time (1990 - early 2020s)	+/- Cyclical, no evidence of secular increase	+/- No evidence for 'greening'; rather, de-greening	+/- Rising
H1. Environmental policy problems	+/- No evidence	+/- Evidence in CMP, but contradicted by evidence from other data sources	+/- Limited evidence (esp. CMP)
H2. Historic left-right conflict	+/- Evidence present, some evidence for M2.3 (ideology) and M2.2 (radical right)	No expectation, but evidence of an effect	+/- Limited evidence (esp. CMP). Evidence consistent with M2.2 only in CMP data.
H3. Public attitudes on environment	+/- Evidence when GDP omitted	+/- Evidence in CMP and expert survey data when GDP omitted	+/- Limited evidence (esp. VAA, CMP)
H4a. Niche party competition (Greens)	+/- No evidence	+/- No evidence	+/- No evidence
H4b. Niche party competition (RR)	No expectation, but limited evidence for negative effect	-/- Evidence in expert survey data	+/- Evidence (VAA and expert surveys)
Each cell first indicates the direction of the hypothesised relationship: +/- = more/less polarisation on the environment; greater/less salience of the environment; or greener/less green positions on the environment.			

Table X summarises the results. Overall, the polarisation data provides evidence of the politicisation of the environment over time, but data on salience appear more equivocal, suggesting episodic increases and no long-term increase since 1990, rather than the steady or increasing prioritisation of the environment. There is no evidence of sustained 'greening' of party preferences. There is very limited evidence that environmental problems lead to 'greening' or politicisation of party preferences. There is some evidence that historic conflict leads to politicisation and, although there was no theoretical expectation, there is also evidence that it leads to greening. There is evidence for

the role of public opinion in greening when GDP is not controlled for, and some evidence for its role in politicisation. The absence of evidence across all data sources for any positive (pro-environmental) effect of Green parties on other parties policy preferences is striking. There is some evidence for the RR parties influencing other parties to de-politicise the environment and to 'de-green'.

Discussion

What evidence is there for politicisation and 'greening' over time? The evidence on politicisation comes in two quite different parts. On the one hand, the *salience* of the environment to parties is characterised more by cyclicity than by any secular increase over time. This runs counter to widely-held expectations about the development of the environment as an issue: that, having emerged in the 1970s and 1980s, the environment would continue to become more salient, driven by increasingly clear and negative problem information (Green-Pedersen and Wolfe, 2009) and potentially exacerbated by the emergence of the problem of climate change during the period. Nor does it tally with what we know about the public salience of the environment which, although cyclical, has steadily increased as each cycle of salience institutionalised the issue on the agenda (need refs here: either European Commission 2025 or Kenny 2021 [handbook chapter, which I haven't got yet]).

On the other hand, there *is* evidence of increasing *polarisation* on the environment over time across all three measures of environmental policy positions. This finding may be attributable to the way environmental protection is frequently framed as being in conflict with economic growth, particularly as the increasingly far-reaching measures associated with net zero climate policy goals often impose unpopular costs on business and individual citizens (Otteni and Weisskircher, 2022). Or it may be linked to changes in politics more generally, which have reshaped the party politics of the environment. Whatever the causes (which are explored below), growing polarisation indicates that the environment is moving further from its (conceptual) origins as a valence issue and towards more contestation and division.

The picture is less clear cut on 'greening', but overall there is no evidence of a sustained 'greening' of party politics. Salience is highly cyclical, as noted above, and while CMP data shows some fluctuation in positions, the expert survey and VAA positional indicators instead suggest that some de-greening has occurred. By these measures, political parties have not generally adjusted to increasing problem pressure or public salience by adopting greener positions. In summary, parties are not devoting ever-more attention to the environment and there is no strong evidence that they are becoming

greener, but they are becoming ever-more divided on the environment. These trends offer little succour for those hoping that politics would evolve to meet the challenge of environmental crises through paying attention to those crises, developing new positions, and becoming more unified.

*I THINK THE SUMMARY SO FAR WORKS NICELY BUT FROM HERE DOWN IT FEELS LIKE THERE IS A FAIR BIT OF REPETITION OF THE ANALYSIS SECTION IN THE FINDINGS. I'VE TRIED TO MAKE IT CONSISTENT WITH THE NEW FINDINGS [BUT PLEASE CHECK] BUT MORE GENERALLY I'M STRUGGLING TO FIND A WAY TO WORK AROUND THE REPETITION ISSUE - BUT IT MAY BE THAT THE TWO OF YOU THINK I EXAGGERATE THE ISSUE?

UPDATE MONDAY - I'M TRYING A SLIGHTLY DIFFERENT APPROACH WHICH EMPHASISES THE HISTORIC CONFLICT FINDING WHICH SEEMS TO BE ONE OF OUR HEADLINE (AND DISTINCTIVE) FINDINGS AS A WAY OF WORKING ROUND THE REPETITION ISSUE - WHAT DO YOU THINK?

These trends are only part of the story. What has driven variation over time?

One clearcut and important finding is the confirmation of H2: *The stronger the historic left-right conflict, the more politicised is the environment*. Moreover, the evidence shows that the historic structure of conflict is associated with both components of politicisation: the salience of the environment to parties and the degree of polarisation over the issue (albeit for polarisation the strength of that evidence varies across data sources). Overall, our evidence points to an important finding that the relationship between historic conflict and politicisation suggests that the structure of politics drives the structure of environmental politics.

ADD LINKS BACK TO LITERATURE

There is mixed evidence for the other three hypotheses Regarding H1 *The more acute are environmental problems, the more politicised is the environment and the 'greener' are party preferences*, there is no association between environmental problems and the salience of the environment to parties. Yet environmental problems do seem to contribute to greater polarisation among parties, so we can say that the politicisation element of the hypothesis is partially confirmed. By contrast, there is evidence that problems affect party positions on the environment, but the findings are contradictory across our data sources: CMP points to a positive relationship at both system and party levels, while expert survey and VAA data suggest a negative relationship at system level, so the 'greening' element is not confirmed.

ADD LINKS BACK TO LITERATURE

The evaluation of H3, *'The more pro-environmental are public attitudes, the more politicised is the environment and the 'greener' are party preferences'*, is a little more complex. The analysis initially finds no significant relationship between public attitudes and party preferences, but when GDP is omitted from the model then there is some evidence supporting a positive relationship with all the indicators for politicisation and greening. These findings suggest that public attitudes play a role in mediating the effect of GDP on party preferences.

ADD LINKS BACK TO LITERATURE

Our final hypothesis has two components: H4a *'The stronger are Green parties, the more politicised is the environment and the 'greener' are party preferences'* and H4b *'The stronger are Radical Right parties, the more politicised is the environment (polarisation only) and the less green are party preferences (position only)'*. Perhaps our most surprising finding is that there is no evidence that Green party success has any impact on party preferences; indeed, although not significant, most of the coefficients are in the opposite direction to that expected. This finding appears to contribute to the small body of literature that suggests that Green party strength has a counterintuitive effect on the greening of party politics (e.g. Abou Chadi 2016). By contrast, our findings contribute new analysis on the role of radical right parties in environmental politics, with expert survey data showing the expected strong negative effect on positions and the expert survey and VAA both revealing the expected positive impact on polarisation.

ADD LINKS BACK TO LITERATURE

here is a draft of a paragraph on using the 3 datasets which probably should go at the end of the Discussion?:

This article makes an important, and as far as we are aware, original contribution by xxx through the analysis of three different types of dataset - CMP, expert surveys and VAA - in contrast with previous studies that have mostly used just one of these datasets. The datasets do not all play an equal role in our study because we use CMP data across everything – descriptive data and analysis, system and

party levels, and the three indicators of salience, position, polarisation - whereas the expert survey and VAA datasets are inevitably employed selectively due to the more limited range of measures.

Of course, using several datasets inevitably brings complexity to our findings and analysis. In some instances all three sources agree on the same finding, perhaps most notably in revealing how polarisation has risen over our period of study. For several indicators (EG?) all or most of the measures point in the same direction, even if they are not all significant (eg??). Elsewhere, the findings are less clearcut but such complexity doesn't necessarily mean that the findings are contradictory. (eg need example, perhaps if we are able to break down VAA?). Where there are sharp differences in the findings that certainly gives us pause for thought and deters from drawing strong conclusions on those points. However, despite the challenges associated with this broad approach we think our attempt to triangulate data sources is a positive move that is consistent with those scholars who argue in favour of the complementarity use of these approaches (da Silva and Garzia) - DO WE WANT TO ARGUE/CLAIM THIS?

These trends are only part of the story. What has driven variation over time? Is there evidence of a role for environmental problems (H1), historic conflict (H2), public salience (H3) or electoral competition from niche parties (H4) ? And is there evidence for these mechanisms at the party level?

Regarding politicisation there is evidence that the *salience* of the environment to parties is associated with historic conflict (H2) and public salience (H3) (when GDP emissions are omitted), but not with environmental problems (H1). For *polarisation*, there is evidence of a relationship between historic conflict (H2) and polarisation on the environment, albeit the strength of that evidence varies across data sources. This relationship suggests that the structure of politics drives the structure of environmental politics. However, only one of the expected mediators of this effect - a positive effect from Radical Right parties - appears to (?) mediate the effect of historic conflict, which further suggests a relationship between the salience of the environment and polarisation on the environment (perhaps cite Abou Chadi et al. 2020 in WEP). There is some evidence that public salience (H3) is associated with polarisation on the environment, especially when GDP is excluded. There is little evidence of niche party influence on politicisation (environmental salience or polarisation), although there is good evidence that a stronger radical right leads to more polarisation on the environment.

As in the analyses of system-level politicisation over time, this gives a differentiated picture of politicisation: evidence of left-right ideology playing a significant role at the party level (in relation to

salience), and evidence of left-right polarisation shaping environmental polarisation. Overall, we can say that left-right politics shapes environmental politics, which supports the ‘incorporation’ perspective.

*

What drives ‘greener’ politics? The evidence for salience is summarised above. On position, there is sharply contradictory evidence on whether environmental problems (H1) drive greener positions: the CMP data seems to confirm this hypothesis, while the expert survey and VAA data find a negative relationship. GDP is positively associated with greener positions at system level across all data sources, but once it is removed public opinion (H3) becomes significant which suggests **WHAT?**

As in the analyses of salience, left-right positions are associated with greener positions. Niche party strength (H4, also M2.2) is remarkable for its failure to produce coefficients in the expected direction, and for sometimes producing coefficients in the opposite direction to that expected. This appears to contribute to the body of findings that suggests that Green party strength has a counterintuitive effect on the greening of party politics (e.g. Abou Chadi 2016), and adds to this some new analysis on the role of radical right parties in environmental politics, albeit these parties also appear, if anything, to have an effect that is counter to expectations.

Finally, although it was not expected, it turns out that there is an association between historic conflict and greener positions, at least in CMP data.

Once again, the role of left-right politics and public salience appears to play a role in ‘greening’. Niche parties do not have the expected effects.

*

I WONDER IF WE MIGHT TRY MORE EXPLICITLY TO STRUCTURE SOME OF THIS SECTION AROUND OUR STATED ABSTRACT CONTRIBUTIONS:

First, what do the findings tell us about the nature of the environmental issue (the valence/positional debate) by providing a system-level description of the structures of competition on the environment, drawing on a range of data sources. [I THINK WE DO THIS THE FIRST BIT OF THE DISCUSSION - I WILL MAYBE ADD ONE OR TWO MORE REFS TO TIE BACK TO THE LIT REVIEW.]

Second, it revisits existing explanations for the politicisation of the environment, contributing to existing research on the roles of niche parties, public attitudes and environmental problems. **[need to add in the link backs to early part of paper]**

Third, it tests a new argument, which suggests that the historic structure of conflict underpinning a party system shapes the politicisation of the environment, and it finds support for this argument.

[need to make this a bit more explicit in the discussion]

Fourth our original contribution using 3 data sources [can build this around the points listed below - but maybe others?]

Potential sources of some of the more puzzling results:

- environment vs economy construct compared to VAA statements (vs prices, vs environmental taxes)
- omitted variables (though we've covered a lot)

Takeaway: vote-seeking matters (but not niche party competition) and historic conflict does drive

Limitations:

[We have not measured some variables directly, e.g., the issue composition of the environmental issue (over time especially), i.e., has the rise of climate made the environment more polarised?]

(see Gingrich & Giudici, 2025 for an example focusing on education).

Our explanatory analysis focus mainly on Western Europe. The nature of environmental politics is quite different in CEECs. (Is this an argument for reporting/presenting it separately in the descriptive analyses too?)

Conclusion

First study to use each of these data sources. Advantages, but also complexity.

Pros and cons of the different data sources

Can 'environment' and 'climate change' be viewed together when it comes to valence? Or, are there differences?

Note from previous draft on politicisation (M2.1) expectations

		Outcome		
Causal factor: mainstream party competition indicated by lagged systemic:		Salience	Polarisation	Position
	Salience t-1	+	+	NA
	Polarisation t-1	+	+	NA

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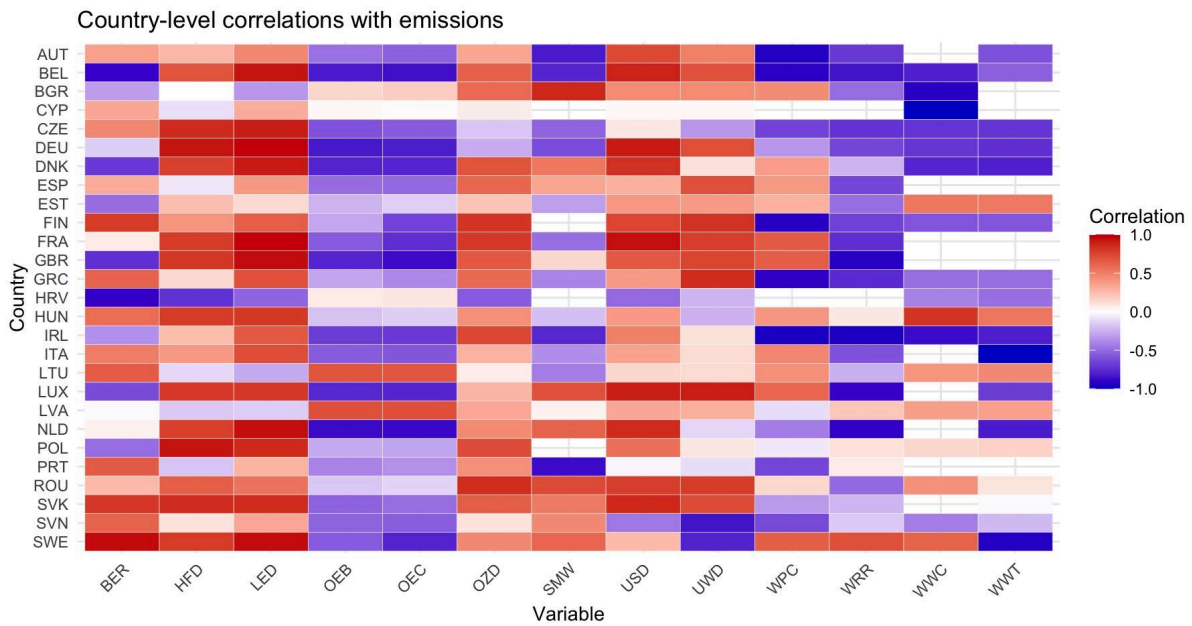
Notes

Journals: EP (8,000 incl. references; max 9,000), EJPR (10,000 words excl. references, abs, appendix – 2 self-citations, max 3 instances), BJPS (10,000 excl. references), CPS (11,000 not including references), PP (8,000 incl. refs), WEP (special issue with Fay), G&O (10,000), Political Studies (9,000 excl. refs – published CGP/CL article). Current aim: keep it tight e.g., 9,000 incl. refs, if possible.

Appendix

GHG emissions and environmental problems

Problem Indicator



Moderate to strong correlations (either negative or positive for many of the indicators)

Indicators:

BER: The Bioclimatic Ecosystem Resilience Index measures the capacity of natural ecosystems to retain species diversity in the face of climate change -

HFD: Measures household solid fuels using the number of age-standardized disability-adjusted life-years lost per 100 (Household air pollution from solid fuels) +

LED: Measures lead exposure using the number of age-standardized disability-adjusted life-years lost per 100 +

OEB: As a proxy of ozone pollution effects on biodiversity +

OEC: As a proxy of ozone pollution effects on crop productivity +

OZD: Measures ozone exposure using the number of age-standardized disability-adjusted life-years lost per 100 +

SMW: Controlled solid waste refers to the percentage of household and commercial waste generated in a country that is collected and treated in a manner that controls environmental risks. This metric counts waste as “controlled” if it is treated through recycling +

USD: Measures unsafe sanitation using the number of age-standardized disability-adjusted life-years lost per 100 +

UWD: Measures unsafe drinking water using the number of age-standardized disability-adjusted life-years lost per 100 +

WPC: Waste generated per capita measures the how much municipal solid waste an average person produces in one year +

WRR: Percentage of wastewater reused after treatment -

WWC: Percentage of wastewater collected for treatment. Sometimes measured as the percentage of the population connected to urban or independent wastewater treatment facilities. -

WWT: Percentage of wastewater that undergoes at least primary treatment. -

Selection of positional variable (CMP)

We choose not to use Lowe's approach of logging the variables for two reasons. First, Lowe et al's (2011) argument for using logs-ratio rests on the assumption that quasi-sentences represent directly opposing poles of the same position (for example in the case of pro- EU integration versus anti- EU integration). In that case, treating one category as the opposing policy position of the other makes a ratio-based diminishing-returns argument intuitive given that each additional negative sentence directly offsets the accumulated weight of positive ones. In our case, this argument is less persuasive, given that per410 and per501 are not opposing poles in policy and parties usually support policies that combine both. Because of this, applying log transformations might risk overstating a structure of opposition onto what is typically a trade-off in emphasis. Second, the log-ratio is highly sensitive to zeros, and the CMP environmental categories frequently contain zeros, especially for small parties, earlier elections, or low-salience contexts. In such cases, log ratios require the addition of arbitrary continuity corrections (like adding a constant such as 1 or 0.5) and when outcomes can be zero, log-based quantities become scale-dependent, such that effects can be made arbitrarily large or small simply through rescaling (Chen and Roth, 2024).

Table 2 sets out the variety of manifesto-based indicators used in the literature on parties' environmental policy preferences. [MORE TEXT HERE to justify our choice]

Table 2. Indicators used in broad cross-national studies that aim to explain variation in parties environmental policy preferences using statistical models

Publication	Scope (issues)	Primary dependent variable	Secondary dependent variables (incl. robustness tests)
Spoon et al. 2014	Environment	Salience (CMP per501)	
Abou-Chadi 2016	Environment	Salience (CMP per501 + per410) and Position (CMP per501 - per410)/(per501 + per410)	Salience (per501 only) and primary DVs (Salience and Position) with per416 also included*
Abou-Chadi et al. 2020	Environment	Position ($\log(\text{abs401} + \text{abs410} + \text{abs703} + 0.5) - \log(\text{abs501} + \text{abs416} + 0.5)$ (abs = absolute number of sentences)	Note also that salience is an IV, measured as CMP per501 + per416 (not logged – they argue against this)

Farstad 2018	Content explicitly related to climate change	Salience (subset of CMP per501)	
Green-Pedersen 2019	Environment	Attention (salience): CAP 700s major topic and several other CAP subtopics)	
Schwörer 2024	Content explicitly related to climate change	Salience (dictionary-based)	Position as salience (positive) minus salience (negative)
Jahn 2016	Environment ('Green vs Growth')	Position: multi-dimensional scaling of four CMP items (408, 410, 411, 501, 416)	
Note: includes studies of environment or climate change, and not studies of party preferences on animal welfare or GMOs, or studies that focus on variation within party families (Schwörer et al., 2023; Schwörer & Fernández-García, 2024b, 2024a). Meguid (2005; 2008) uses per501 + per416 as an indicator of an accommodative strategy towards Green parties and per401 + per703 + per109 as an indicator of an adversarial strategy towards Green parties. These are independent variables that she uses to explain Green party vote share. * These are the same components as in Lowe et al.'s (2011) additive scaled dimension on environmental protection, although they also find the log.			

Regression tables

[Appendix Tables 15 April 2026.docx](#)

Interaction plots

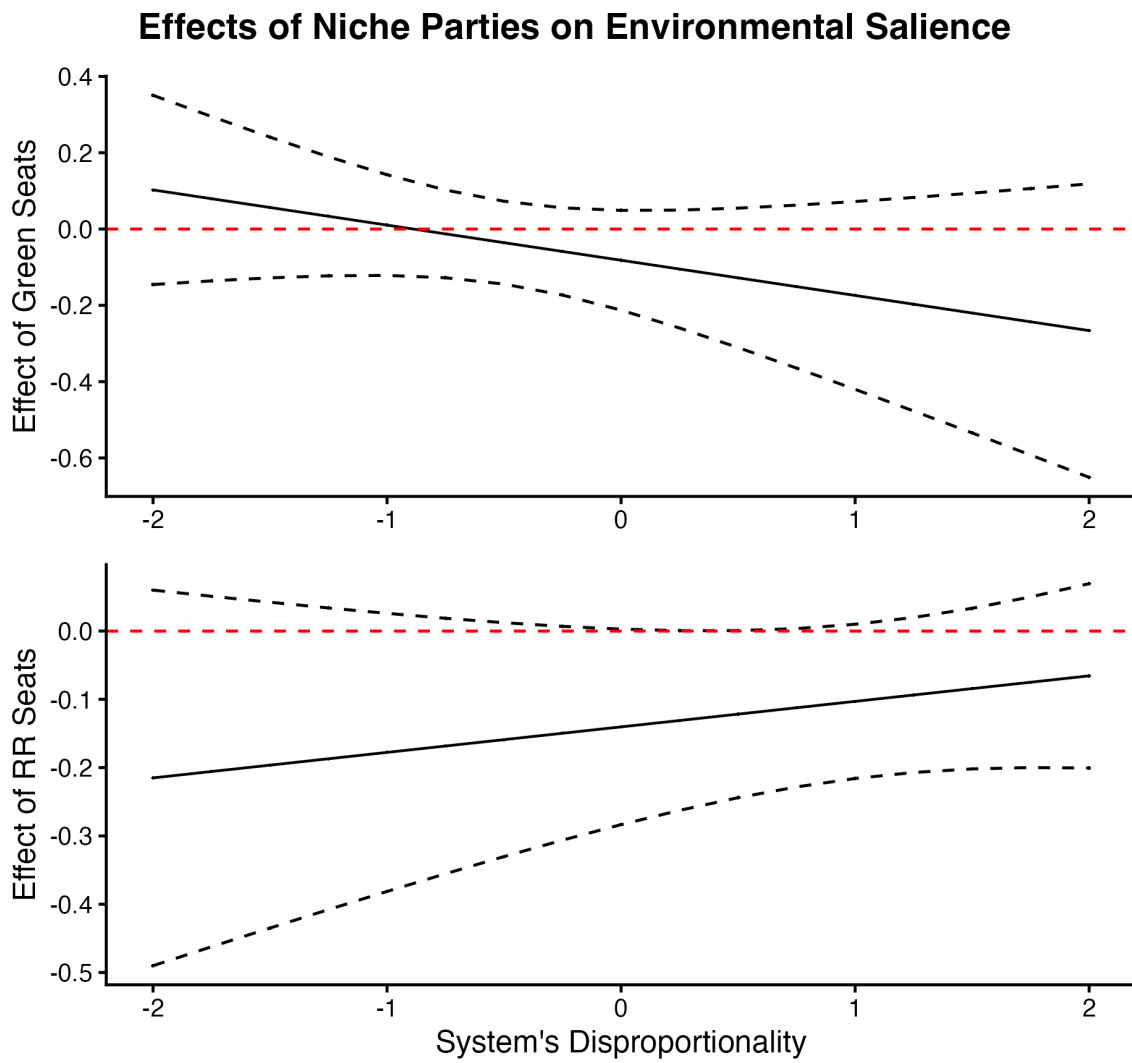


Figure. Effects of niche parties on environmental salience over disproportionality (Appendix Table A4, model 5. CMP data)

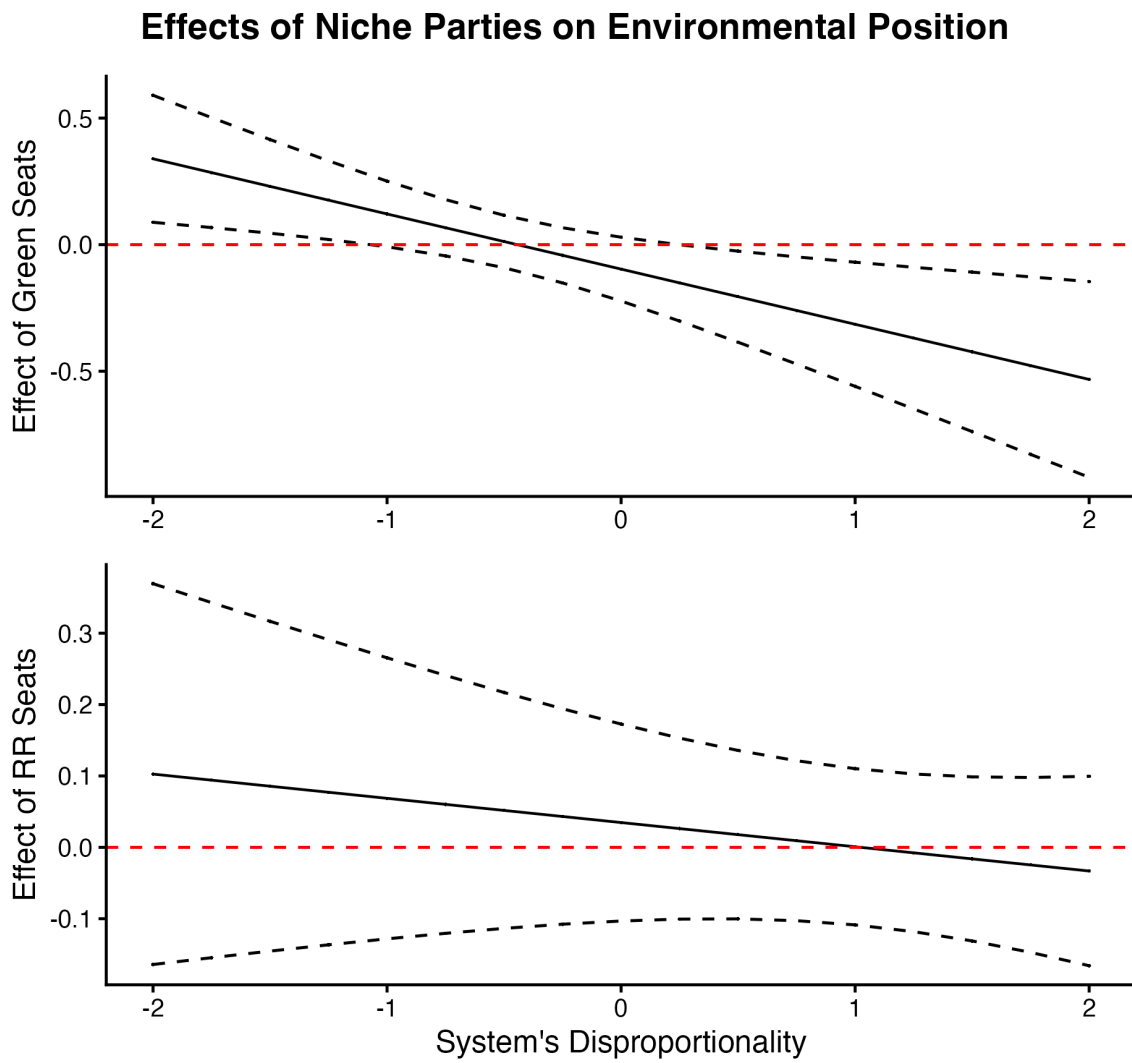


Figure. Effects of niche parties on environmental position over disproportionality (Appendix Table XX, model X. CMP data)

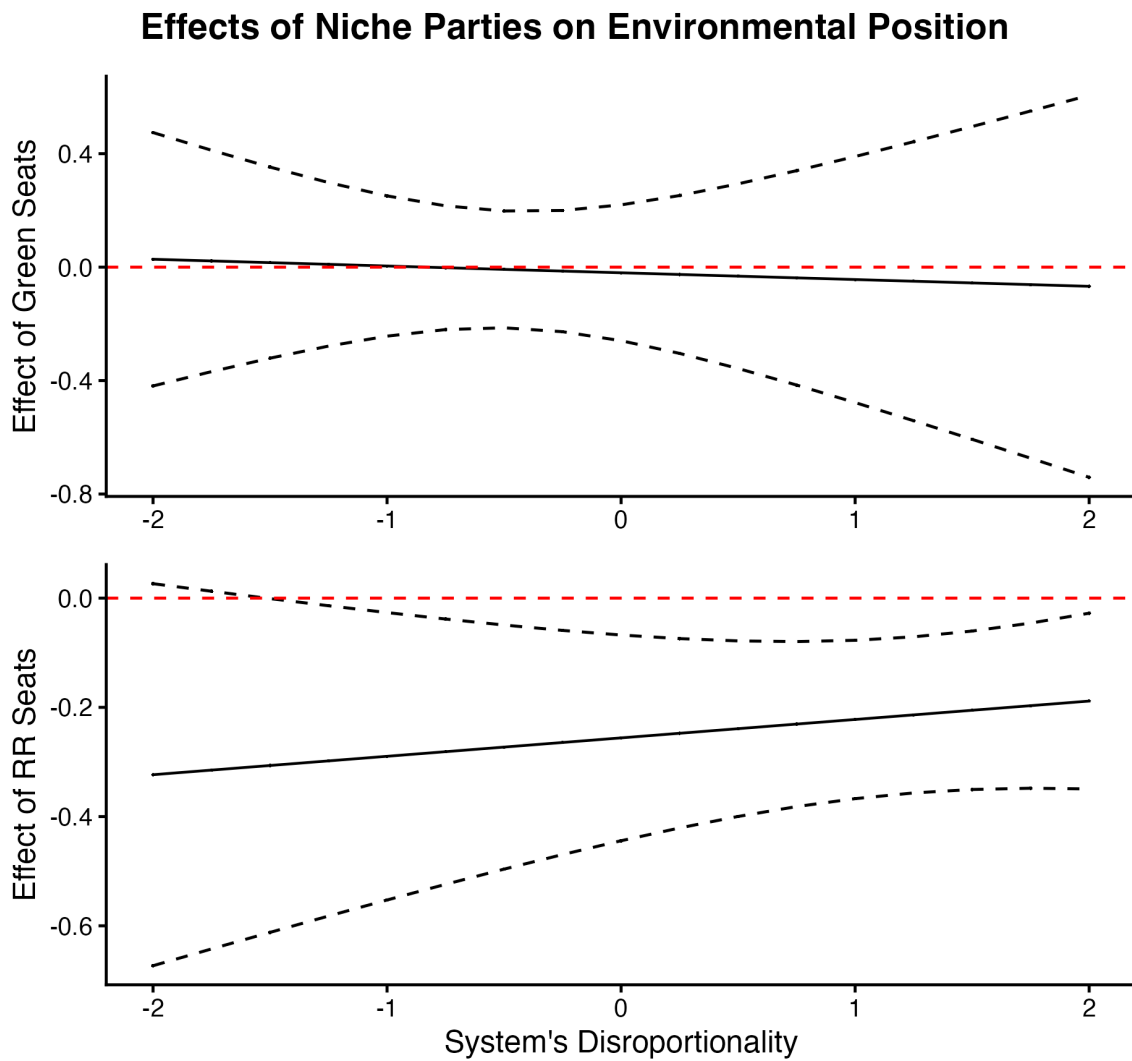


Figure. Effects of niche parties on environmental position over disproportionality (Appendix Table XX, model X. Expert survey data)

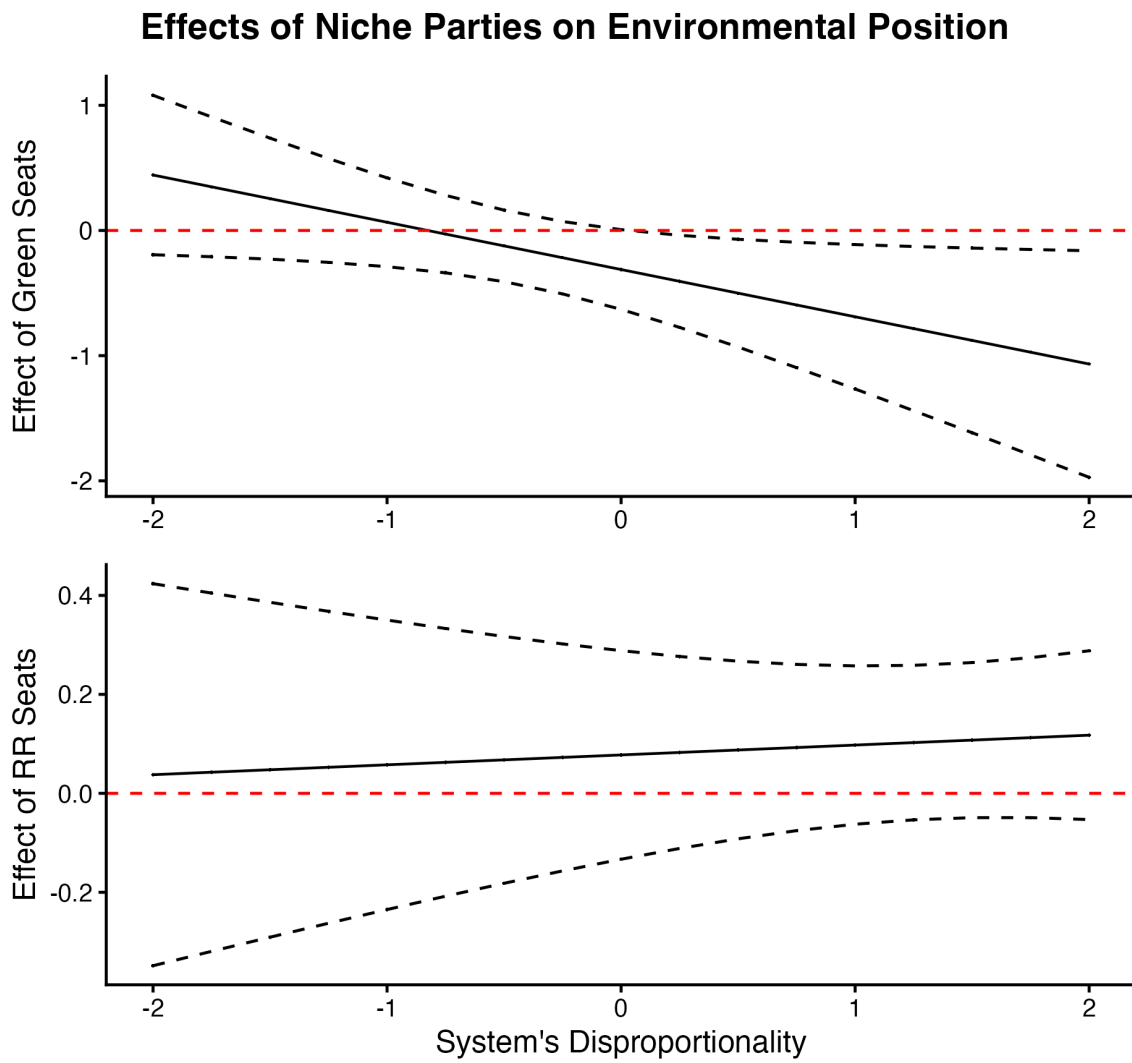


Figure. Effects of niche parties on environmental position over disproportionality (Appendix Table XX, model X. VAA data)

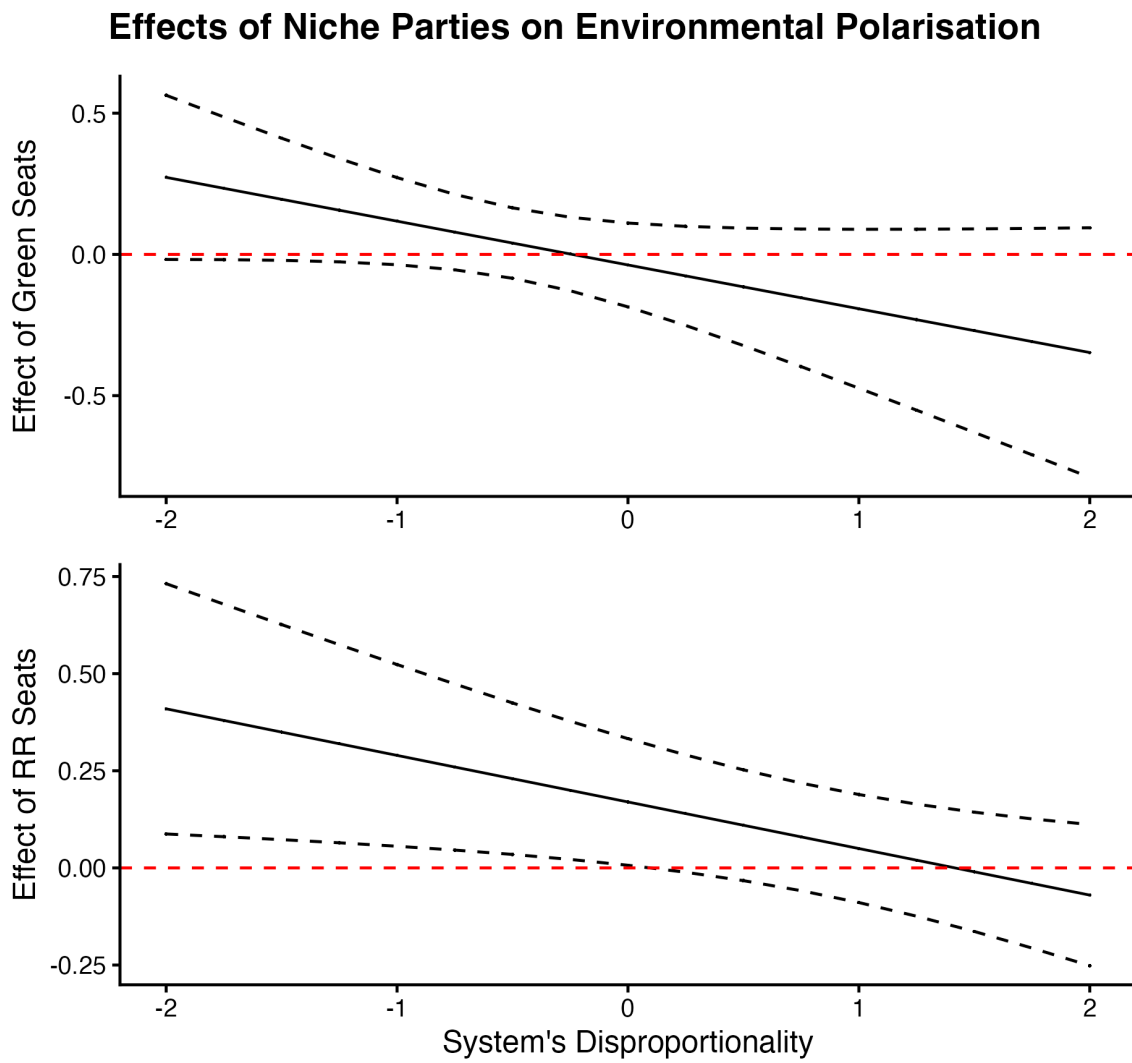


Figure. Effects of niche parties on environmental polarisation over disproportionality (Appendix Table XX, model X. CMP data)

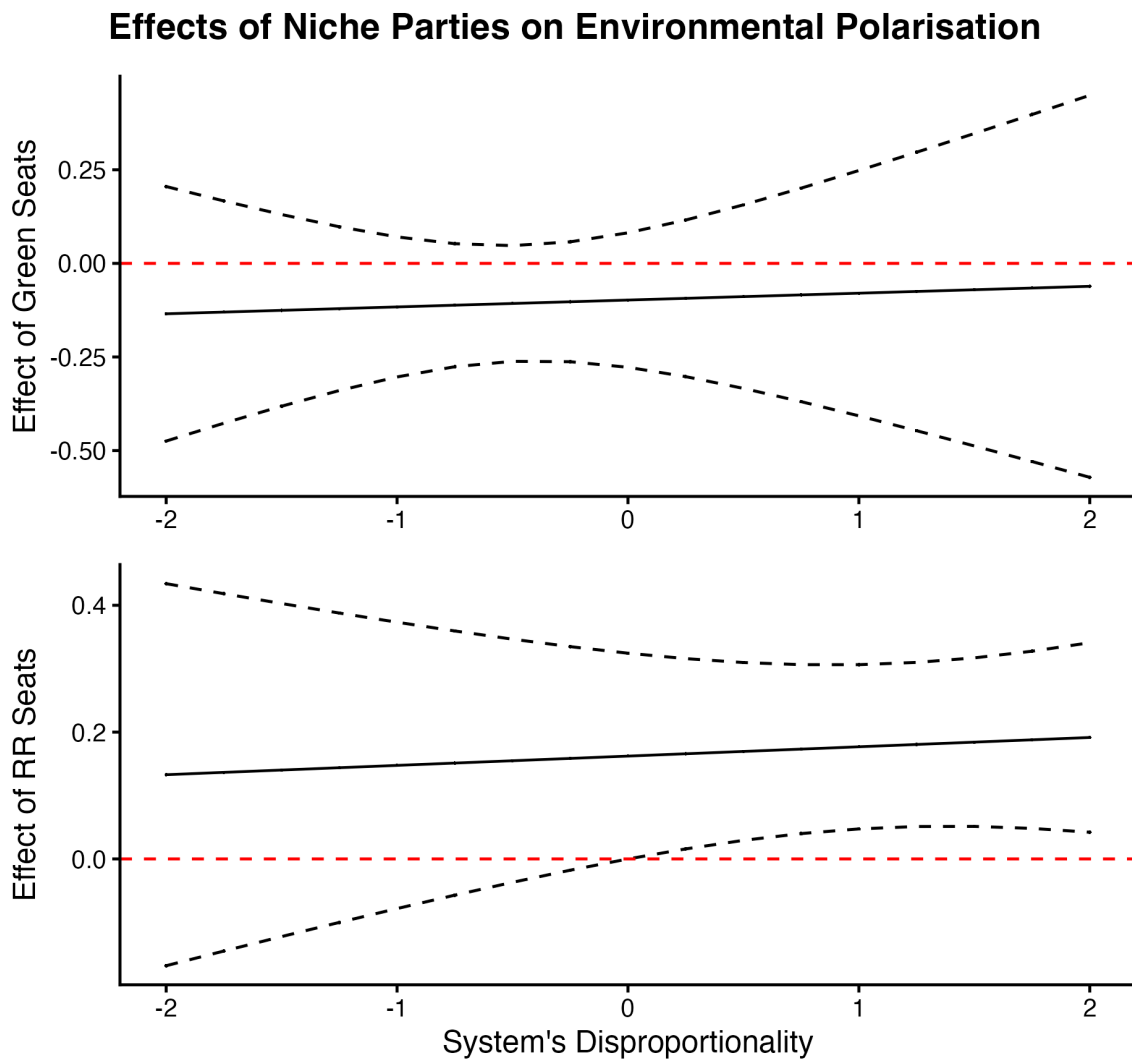


Figure. Effects of niche parties on environmental polarisation over disproportionality (Appendix Table XX, model X. Expert survey data)

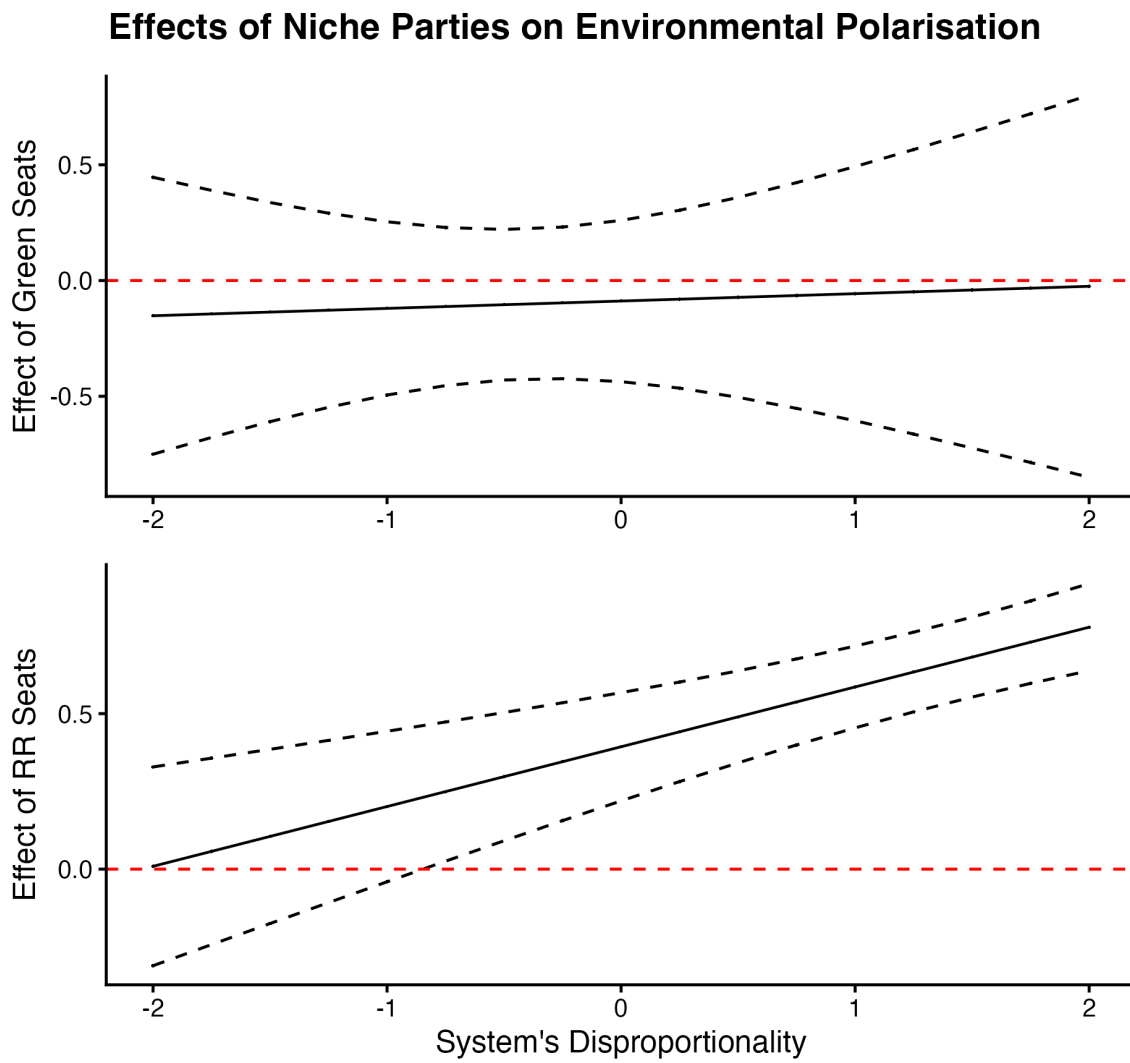


Figure. Effects of niche parties on environmental polarisation over disproportionality (Appendix Table XX, model X. VAA data)

